

Code Generation

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Topic:

Code Generation

Section:

Global Register
Allocation

Managing Registers
Across Calls

Registers Usage in
scip

Instruction Selection

Typical Front Ends

Parser



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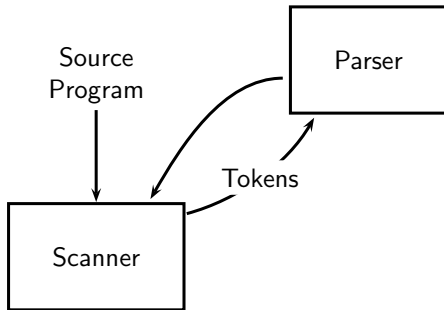
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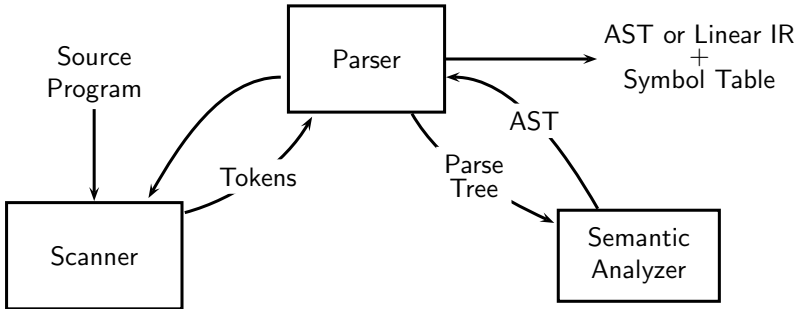
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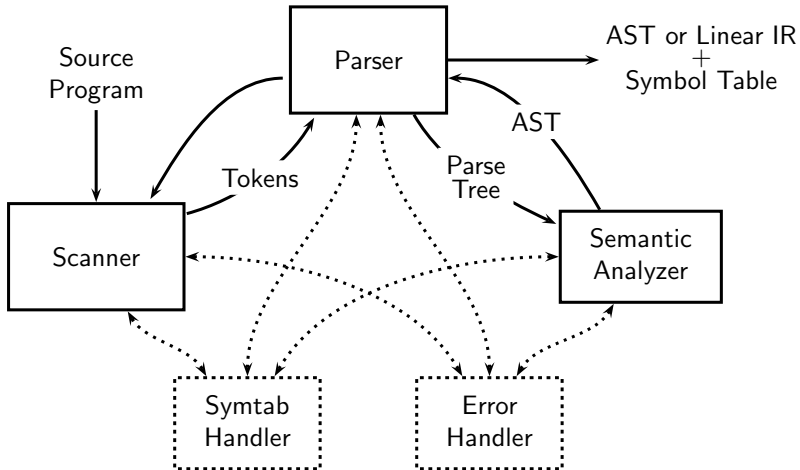
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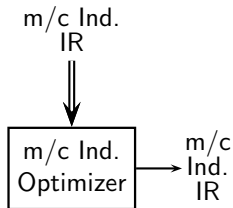
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Typical Back Ends in Aho-Ullman Model



- Compile time evaluations
- Eliminating redundant computations



Typical Back Ends in Aho-Ullman Model

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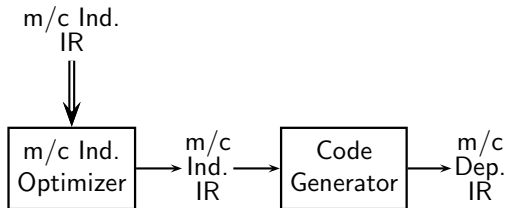
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Instruction Selection



- Compile time evaluations
- Eliminating redundant computations
- Instruction Selection
- Local Reg Allocation
- Choice of Order of Evaluation



Typical Back Ends in Aho-Ullman Model

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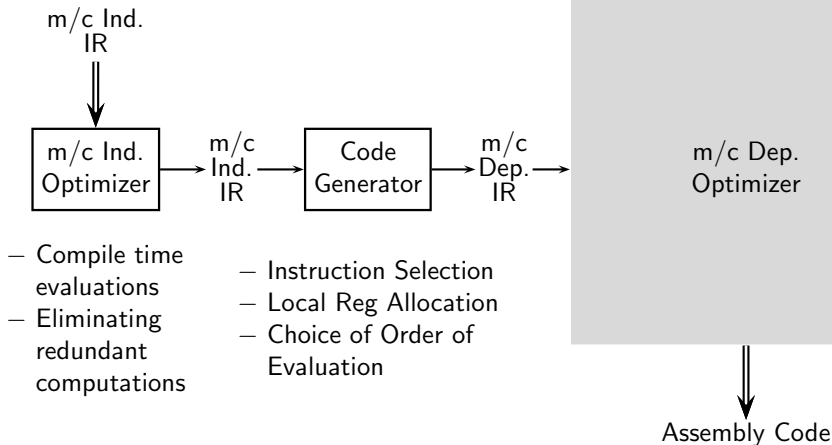
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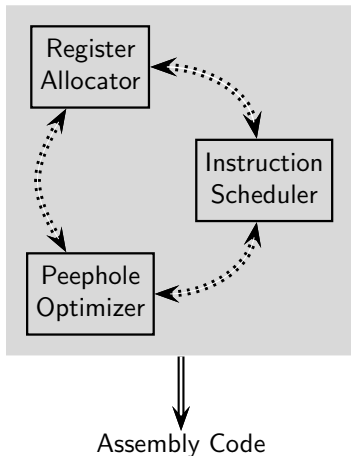
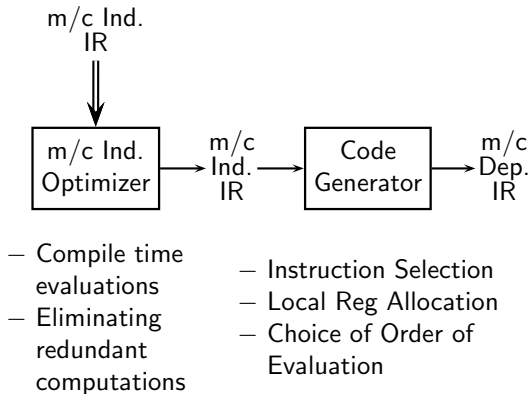
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Instruction Selection

Code Generation

- Register Allocation
- Instruction Selection
- Instruction Scheduling
- Peephole optimization

We will cover this

We will cover this



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Global Register Allocation



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Instruction Selection

Register Allocation

- Accessing values from registers is much faster than accessing them from memory
Latencies for Intel Haswell in terms of cycles
Register 1, L1 cache 4/5, L2 cache 12, L3 cache 36, RAM 264



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Register 1, L1 cache 4/5, L2 cache 12, L3 cache 36, RAM 264
- Issues
 - The number of registers is very small and the number of variables is large
 - Which variables should have their values in registers?
 - In which region of the program should the values be kept in registers?



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Register 1, L1 cache 4/5, L2 cache 12, L3 cache 36, RAM 264
- Issues
 - The number of registers is very small and the number of variables is large
 - Which variables should have their values in registers?
 - In which region of the program should the values be kept in registers?
- Categories
 - Local register allocation
 - Using registers to hold intermediate values of expressions
 - Usually, instruction selection algorithms handle this
 - Global register allocation
 - Keeping the values in registers across statements

We will cover this

We will cover this

Global Register Allocation Using Graph Colouring



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Instruction Selection

Most popular approach

- - Identify live ranges
 - Construct interference graph
 - Colour the graph

Global Register Allocation Using Graph Colouring



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Most popular approach

- - Identify live ranges
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- NP-complete in general
 - Excellent heuristics exists
 - We will study Chaitin-Briggs allocator

Global Register Allocation Using Graph Colouring



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Most popular approach

- - Identify live ranges
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 - Colour the graph
- NP-complete in general
 - Excellent heuristics exists
 - We will study Chaitin-Briggs allocator
- Decidable for chordal graphs

Every cycle of length 4 or more has a chord connecting two nodes with an edge that is not part of the cycle (applies recursively)

 - Most practical interference graphs are chordal
 - All interference graphs for SSA representation are chordal



Motivating Example for Register Allocation

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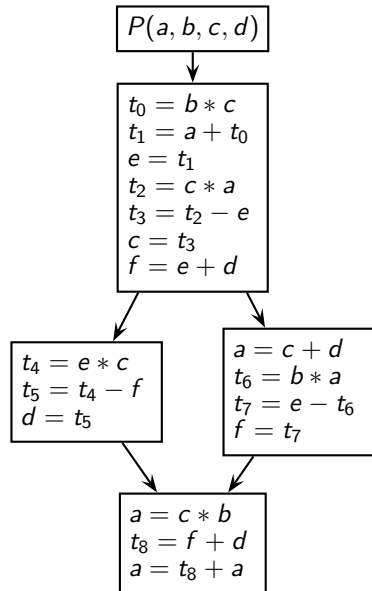
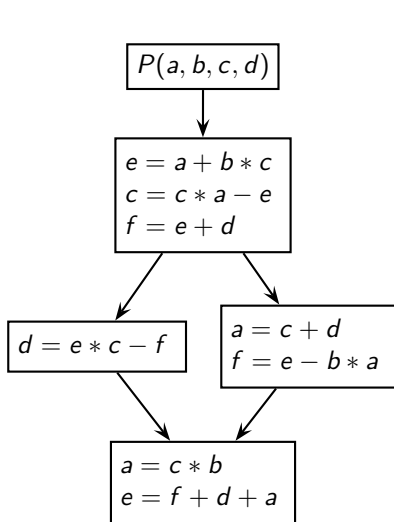
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Steps in Chaitin-Briggs Register Allocator

1. Coalescing

- Eliminating copy statements $x = y$ so that same register can be used for both x and y
- We use copy propagation optimization for the purpose that replaces uses of x by y and eliminate the copy statement $x = y$

2. Identification of live ranges

- Sequences of statements from a definition of a variable to its last use of that value
- We use live variables analysis for the purpose

3. Identification of interference and construction of interference graph

Live ranges l_1 and l_2 interfere if a definition of the variable of l_1 occurs in l_2 or vice-versa

4. Simplification of interference graph to identify the order in which the nodes should be coloured

5. Colouring the nodes



Copy Propagation Optimization (1)

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Instruction Selection

- Perform reaching definition analysis
 - Assignment $n : x = RHS$ gives rise to definition x_n
 - Compute the sets of definition reaching every statement in the procedure
 - If definition x_n reaches assignment $m : x = \dots$, then x_n is *killed* and a new definition x_m is *generated*

When a definition x_n reaches some statement m , it suggests the existence of a control flow path from statement n to statement m along which x is not modified

- Set up the data flow equations over the control flow graph and compute the least fixed point solution
- This amounts to compute the def-use (and use-def) chains in a program



Copy Propagation Optimization (2)

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Instruction Selection

- A use of x in statement m undergoes copy propagation if
 - a single definition $n : x = RHS$ reaches statement m , and
 - RHS is a variable and not an expression, and
 - the RHS variable is not modified along any path from statement n to statement m

The use of x is replaced by the RHS variable

- After copy propagation, the assignment $n : x = RHS$ becomes dead and can be removed



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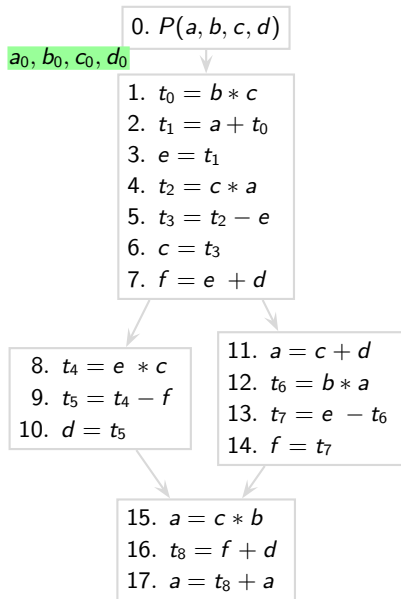
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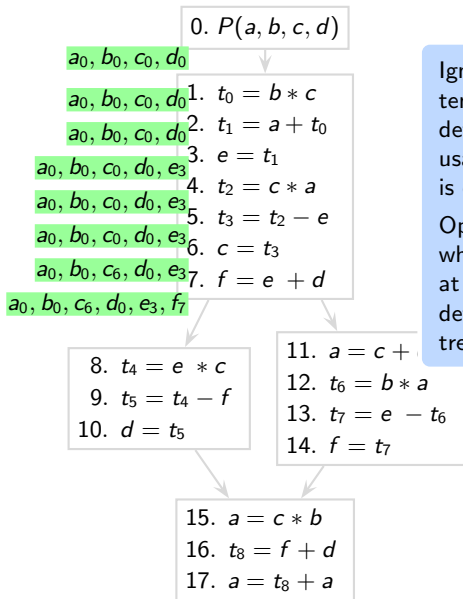
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Copy Propagation Optimization in Our Program



Ignoring the definitions of temporaries because connecting the definitions of temporaries to their usage is trivial because a temporary is defined only once here.

Optimizations may change this in which case, a temporary that is live at the exit of a basic block or is defined multiple times should be treated as a source variable.



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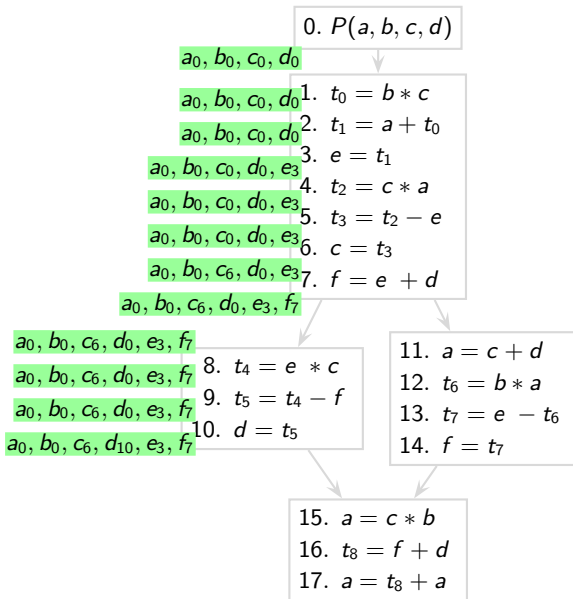
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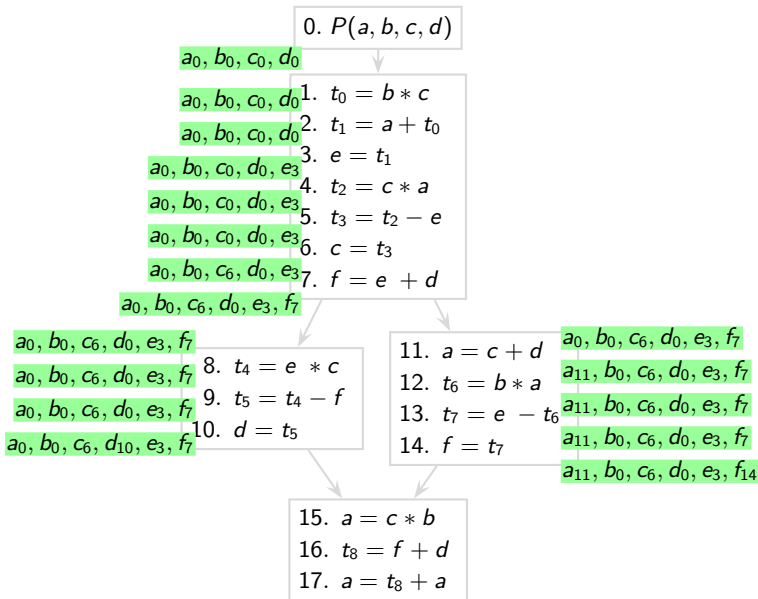
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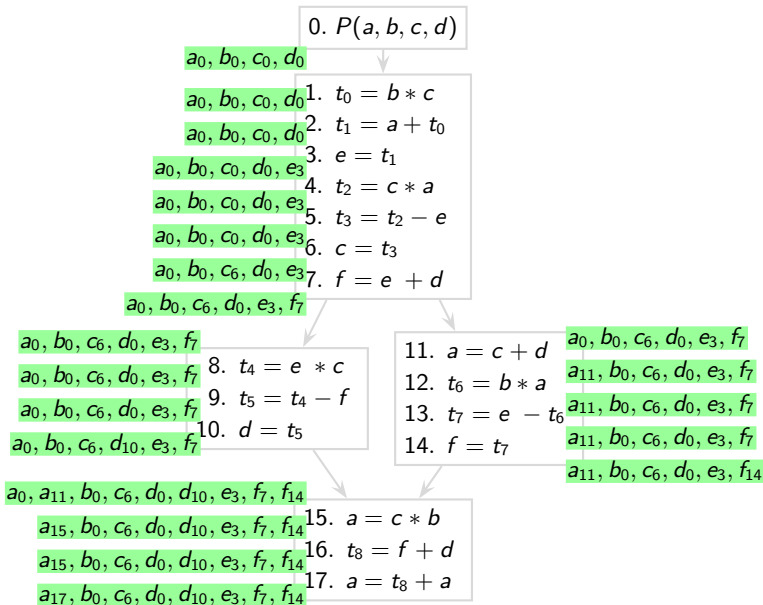
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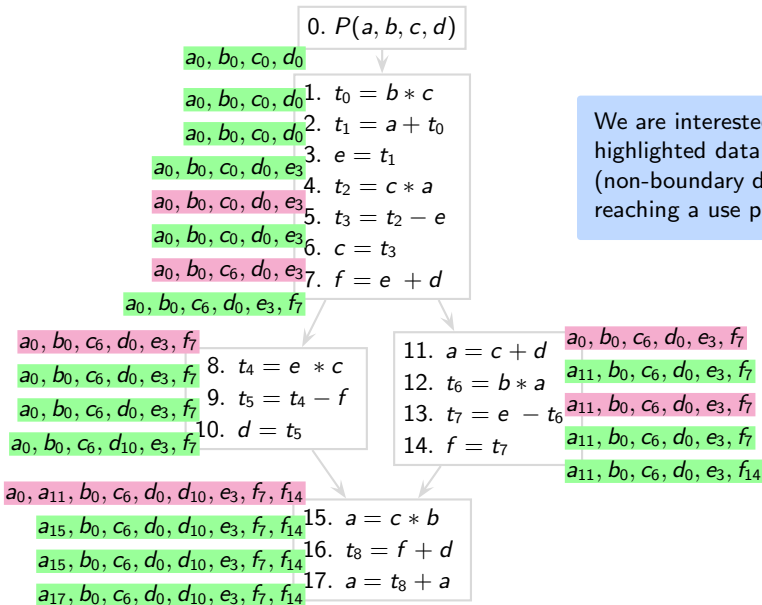
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We are interested in the highlighted data flow values (non-boundary definitions reaching a use point).



Copy Propagation Optimization in Our Program

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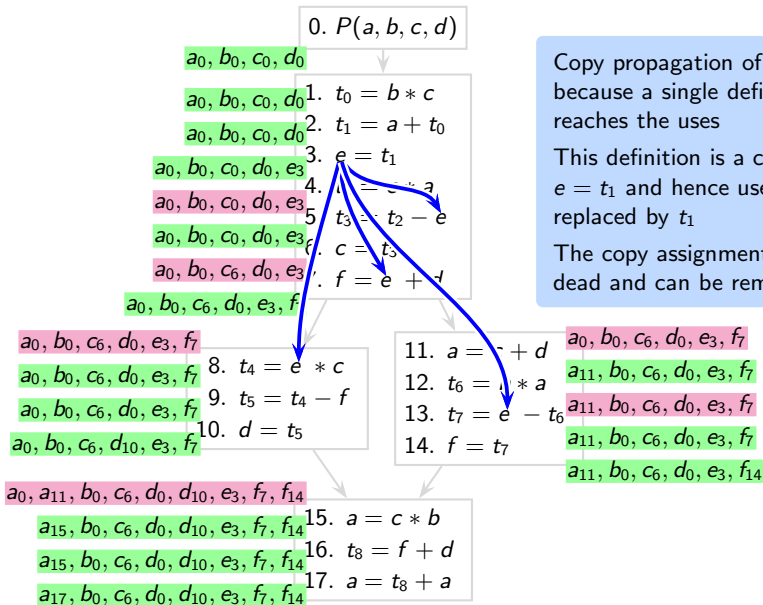
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Copy propagation of e is possible because a single definition of e (e_3) reaches the uses

This definition is a copy assignment $e = t_1$ and hence uses of e can be replaced by t_1

The copy assignment $e = t_1$ becomes dead and can be removed



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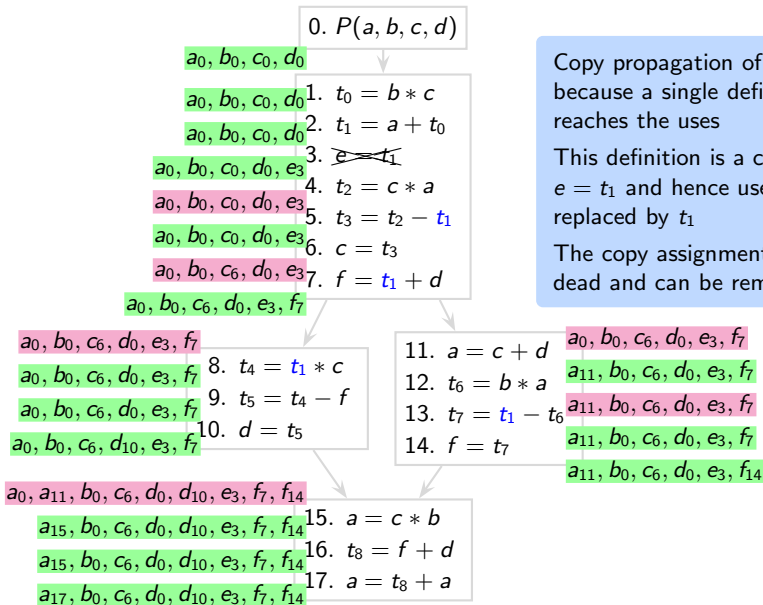
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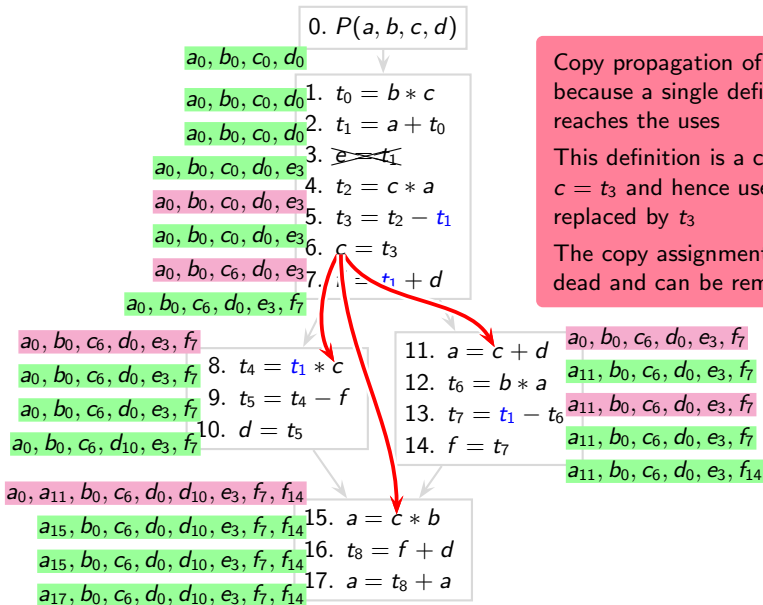
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Copy propagation of c is possible because a single definition of c (c_6) reaches the uses

This definition is a copy assignment $c = t_3$ and hence uses of c can be replaced by t_3

The copy assignment $c = t_3$ becomes dead and can be removed



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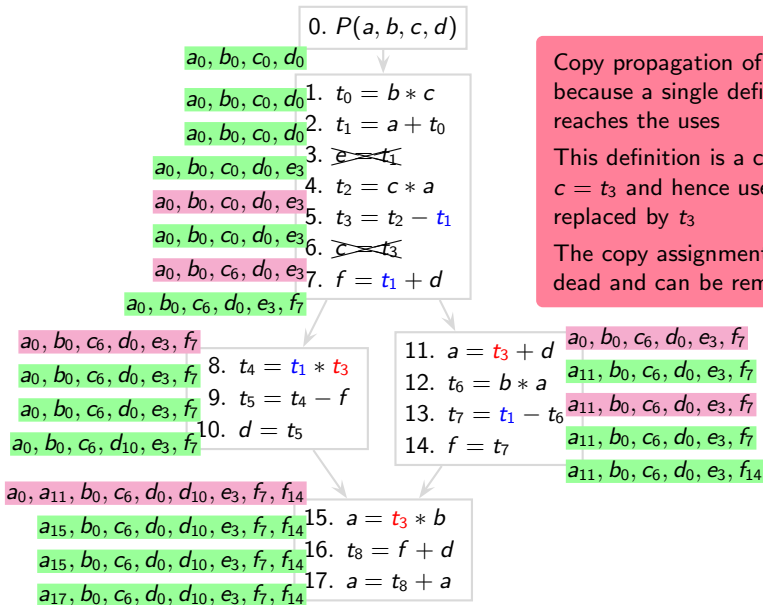
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Copy propagation of c is possible because a single definition of c (c_6) reaches the uses

This definition is a copy assignment $c = t_3$ and hence uses of c can be replaced by t_3

The copy assignment $c = t_3$ becomes dead and can be removed



Copy Propagation Optimization in Our Program

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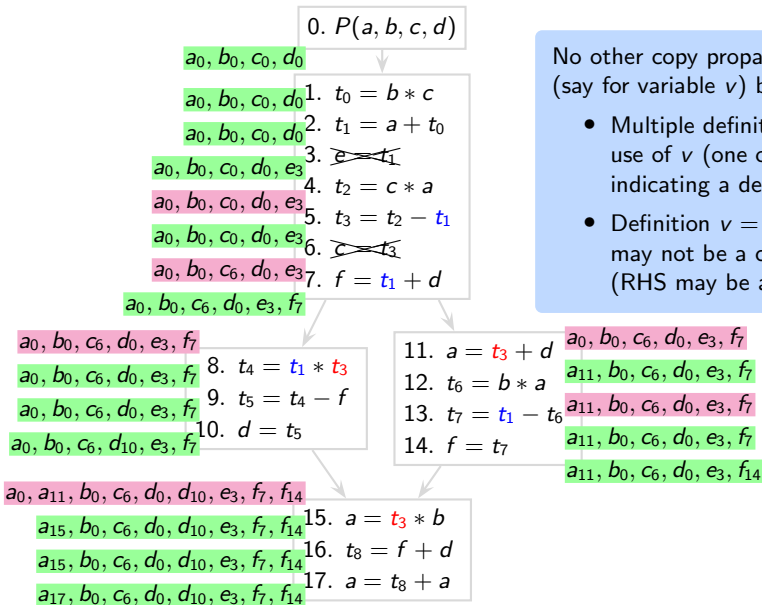
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No other copy propagation can be done (say for variable v) because

- Multiple definitions of v reach a use of v (one of them could be v_0 indicating a definition-free path)
- Definition $v = \dots$ of variable v may not be a copy assignment (RHS may be an expression)



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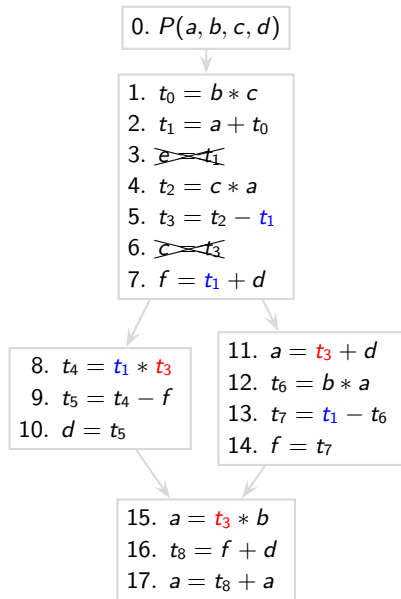
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Instruction Selection

Discovering Live Ranges

- Perform live variables analysis
 - The use of a variable v in statement m makes it live before statement m
 - The definition of a variable v in statement m kills its liveness

If a variable v is live after statement m , it suggests the existence of a control flow path from statement m to EXIT along which x is used before being modified

- Set up the data flow equations over the control flow graph and compute the least fixed point solution
- Statement n is in the live range of variable v if,
 - a definition of v reaches statement n and v is live just before of n , or
 - statement n defines v and v is live just after n



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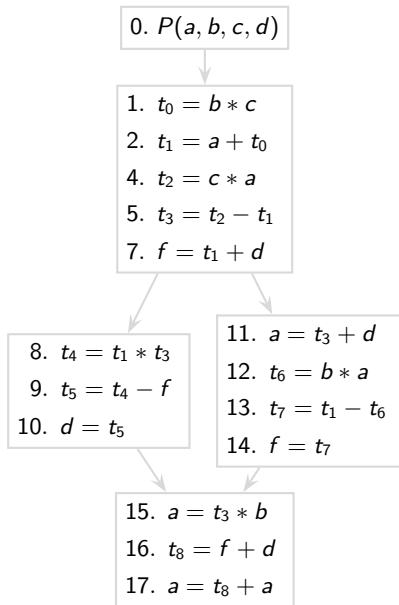
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Discovering Live Ranges in Our Program





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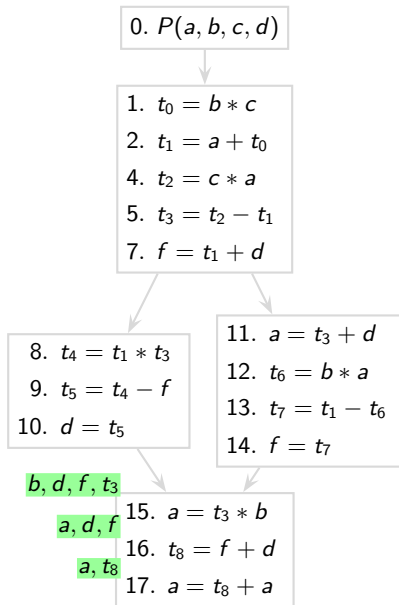
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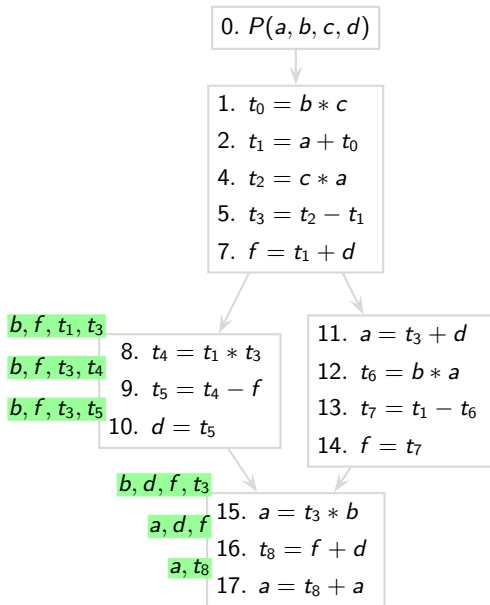
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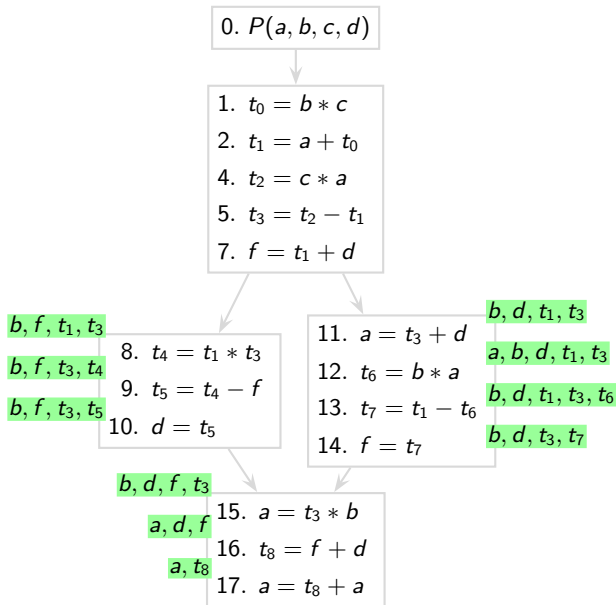
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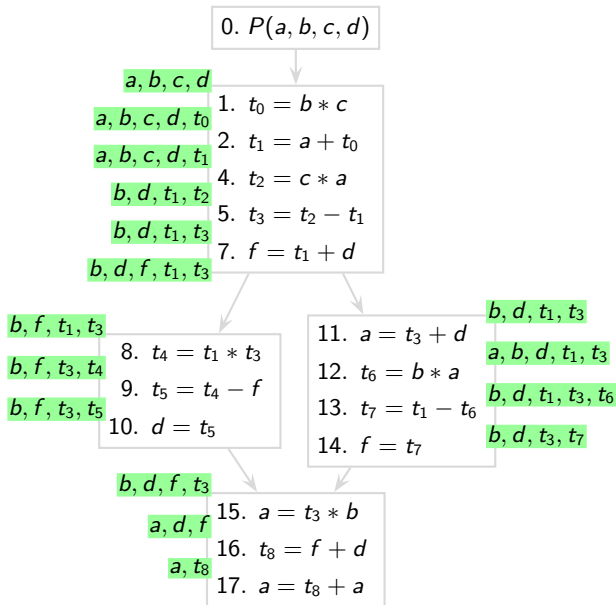
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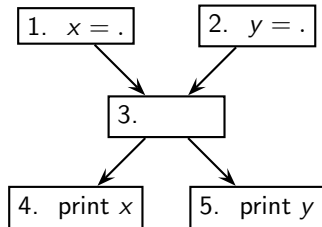




Identifying Interference

- Live ranges l_1 and l_2 interfere if a definition of the variable of l_1 occurs in l_2 or vice-versa

- Consider the example on the right
 - $l_x = \{1, 3, 4\}$ and $l_y = \{2, 3, 5\}$
 - $l_x \cap l_y \neq \emptyset$, yet the same register can be given to both x and y without any problem
 - l_x and l_y do not interfere

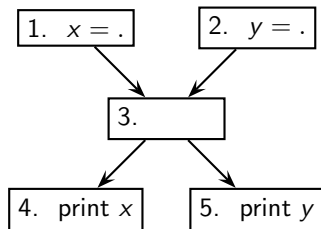


- Both x and y are live at the exit of nodes 1 and 2; however l_x does not include 2 (no definition of x at 2) and l_y does not include 1 (no definition of y at 1)
- Definitions of both x and y reach the entry of nodes 4 and 5; however l_x does not include 5 (x is not live in 5) and l_y does not include 4 (y is not live in 4)



Identifying Interference

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 - l_x and l_y do not interfere
- Both x and y are live at the exit of nodes 1 and 2; however l_x does not include 2 (no definition of x at 2) and l_y does not include 1 (no definition of y at 1)
- Definitions of both x and y reach the entry of nodes 4 and 5; however l_x does not include 5 (x is not live in 5) and l_y does not include 4 (y is not live in 4)
- What would happen if 4 (or 5) had print x, y ?

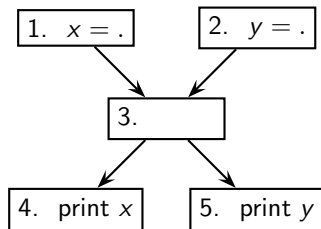




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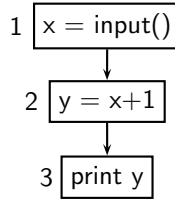


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- Definitions of both x and y reach the entry of nodes 4 and 5; however l_x does not include 5 (x is not live in 5) and l_y does not include 4 (y is not live in 4)
- What would happen if 4 (or 5) had print x, y ?
 - x must be defined on the paths containing 1 and
 - y must be defined on the paths containing 2



Coalescing Beyond Copy Propagation

- Copy propagation eliminates copies only when a variable can be replaced by another variable or a constant



- No copy propagation possible
 - $I_x = \{1, 2\}$, $I_y = \{2, 3\}$ and the definition of y occurs in a statement in I_x
 - However, I_x and I_y can be coalesced because statement 2 is the last use of x and defines y
- If a statement defines a variable (say y) and the statement contains the “last use” of a definition of variable x (i.e., x is live at the entry of the statement but not at the exit of the statement)
then the live ranges of x and y can be coalesced



Constructing the Interference Graph for Our Program

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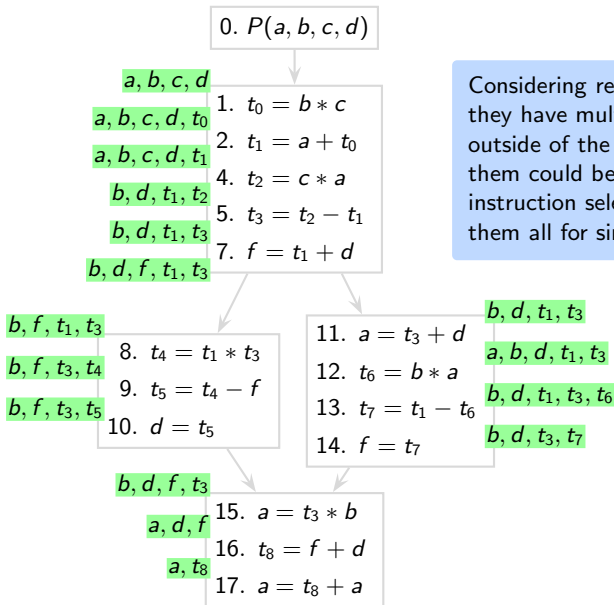
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Considering temporaries because they have multiple uses and some outside of the basic block. Some of them could be handled by instruction selection but we consider them all for simplicity.



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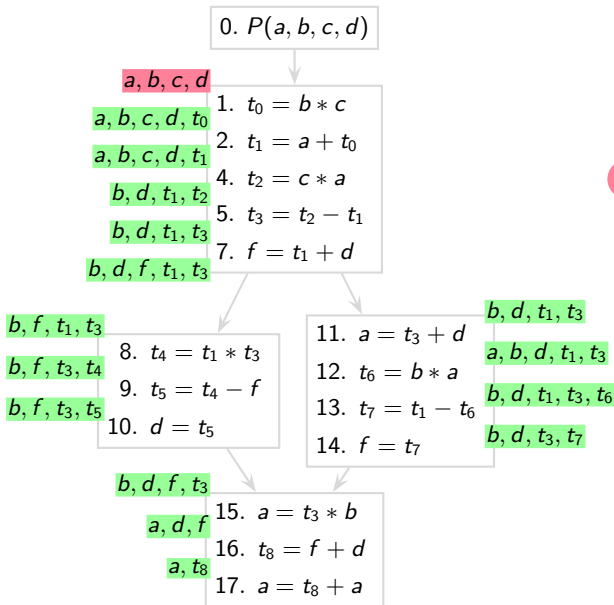
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Instruction Selection

0. $P(a, b, c, d)$

a, b, c, d

a, b, c, d, t_0

a, b, c, d, t_1

b, d, t_1, t_2

b, d, t_1, t_3

b, d, f, t_1, t_3

1. $t_0 = b * c$

2. $t_1 = a + t_0$

4. $t_2 = c * a$

5. $t_3 = t_2 - t_1$

7. $f = t_1 + d$

b, f, t_1, t_3

b, f, t_3, t_4

b, f, t_3, t_5

8. $t_4 = t_1 * t_3$

9. $t_5 = t_4 - f$

10. $d = t_5$

b, d, f, t_3

a, d, f

a, t_8

15. $a = t_3 * b$

16. $t_8 = f + d$

17. $a = t_8 + a$

11. $a = t_3 + d$

12. $t_6 = b * a$

13. $t_7 = t_1 - t_6$

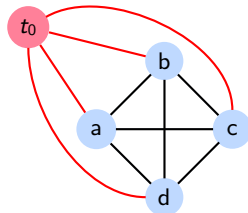
14. $f = t_7$

b, d, t_1, t_3

a, b, d, t_1, t_3

b, d, t_1, t_3, t_6

b, d, t_3, t_7





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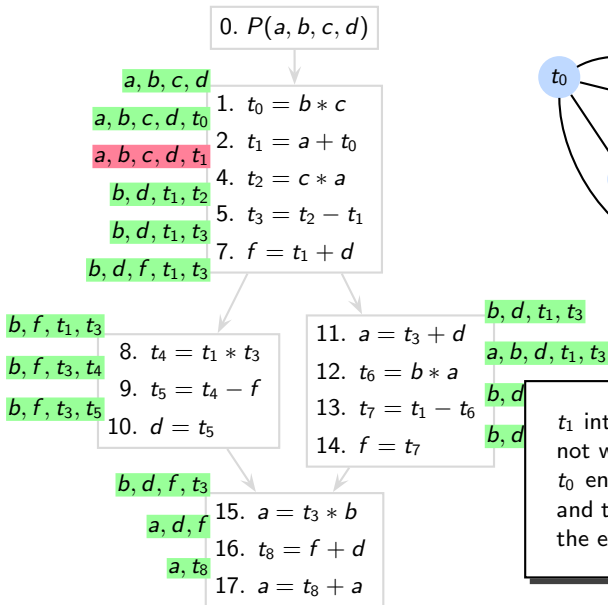
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t_1 interferes with a , b , c , and d , but not with t_0 because the live range of t_0 ends at the entry of statement 2 and the the live range of t_1 begins at the exit of statement 2



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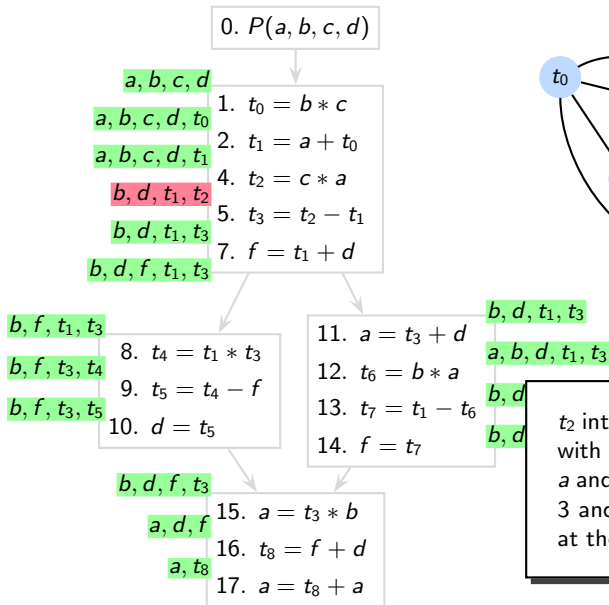
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t_2 interferes with b , d , and t_1 , but not with a and c because the live range of a and c end at the entry of statement 3 and the the live range of t_2 begins at the exit of statement 3



Constructing the Interference Graph for Our Program

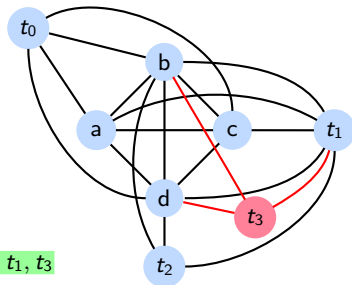
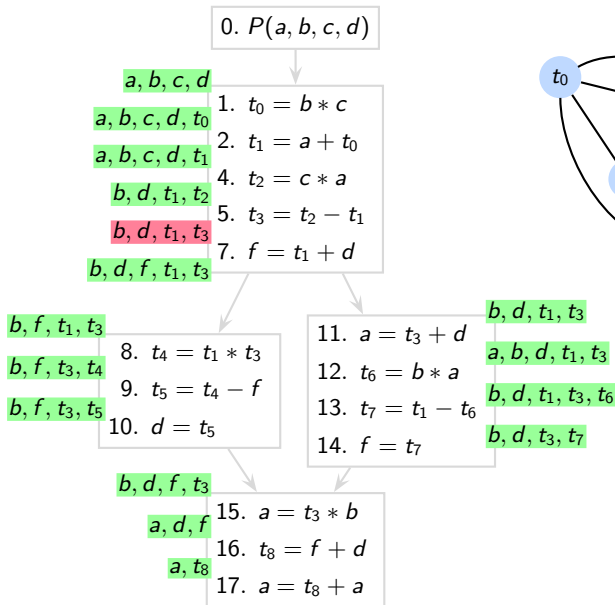
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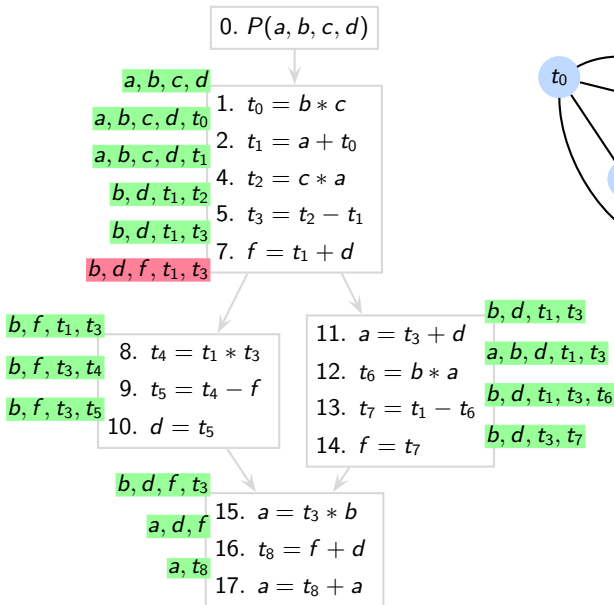
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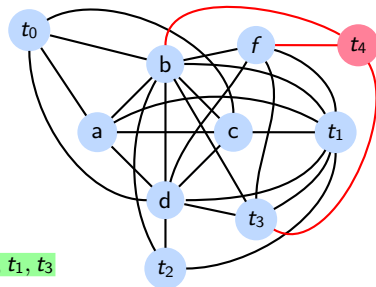
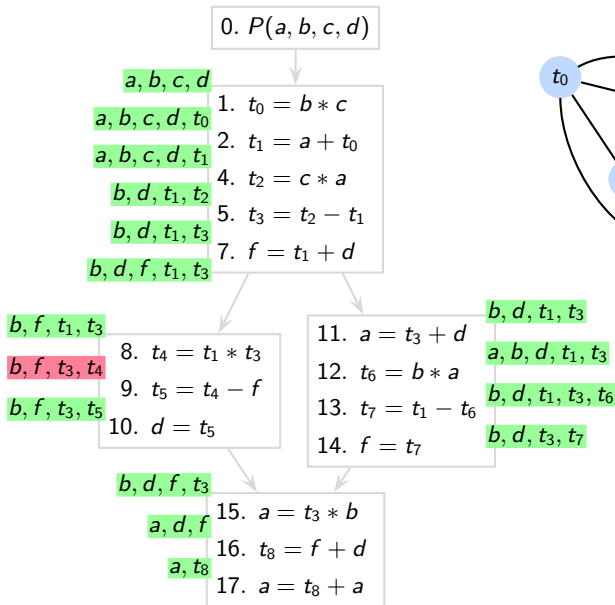
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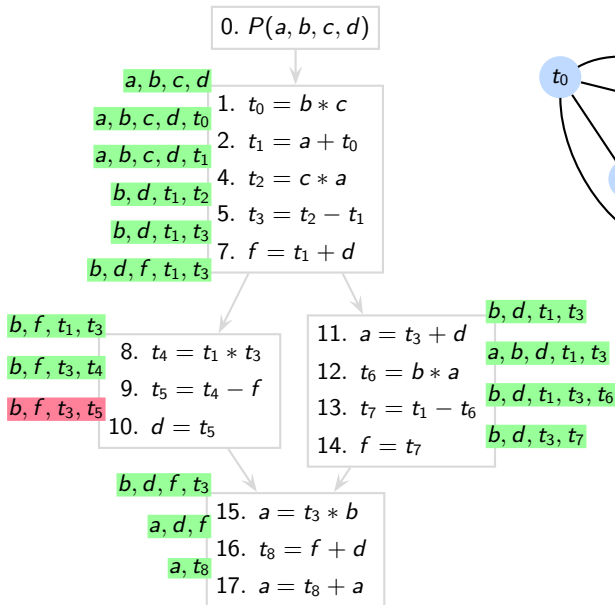
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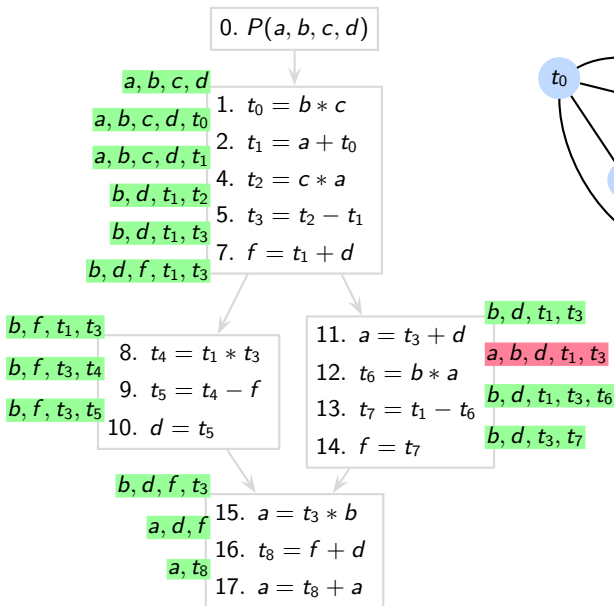
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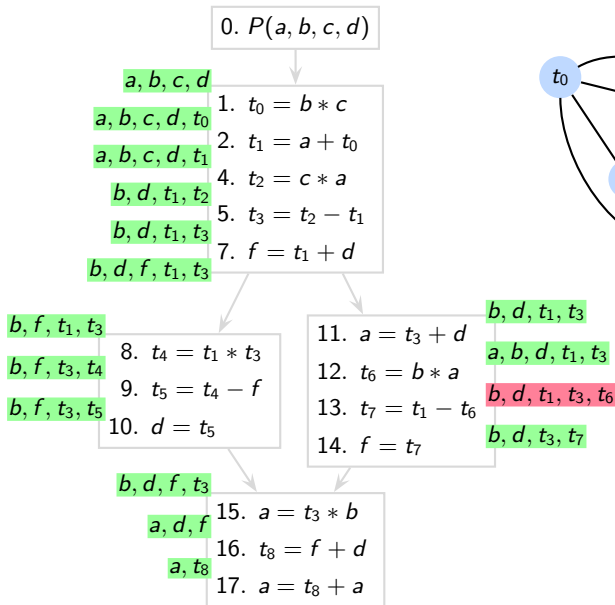
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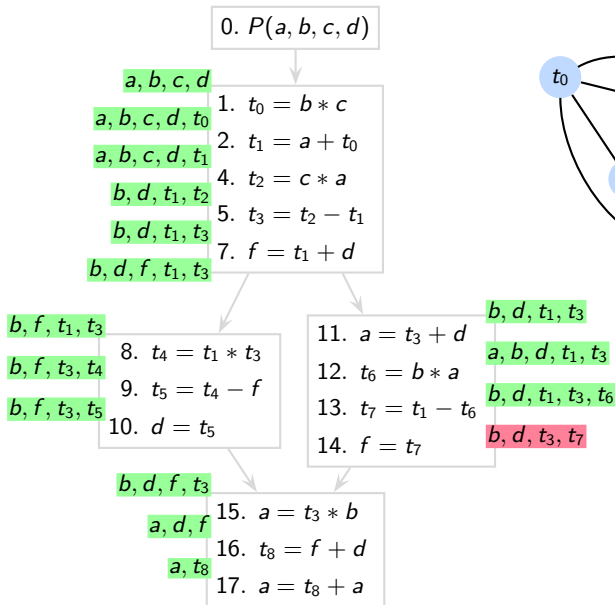
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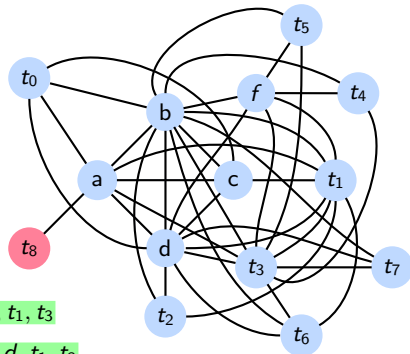
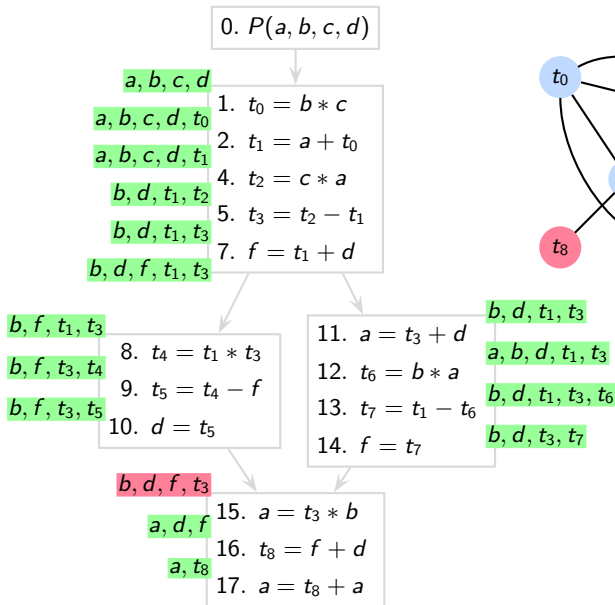
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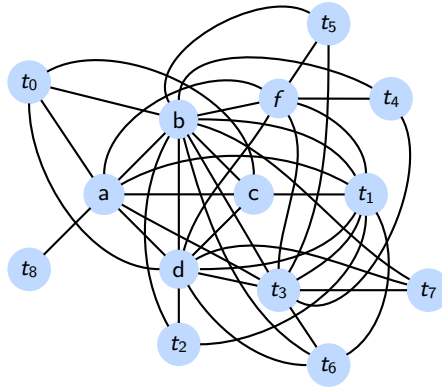
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The Resulting Interference Graph



Node	Degree
<i>a</i>	8
<i>b</i>	11
<i>c</i>	5
<i>d</i>	10
<i>f</i>	6
<i>t</i> ₀	4
<i>t</i> ₁	8
<i>t</i> ₂	3
<i>t</i> ₃	9
<i>t</i> ₄	3
<i>t</i> ₅	3
<i>t</i> ₆	4
<i>t</i> ₇	3
<i>t</i> ₈	1



Live Range Information

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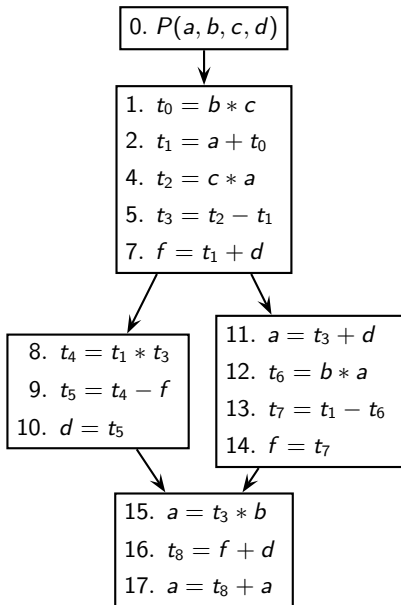
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Live range	Degree D	Loads L	Stores S	Spill cost $C = L + S$
a	8	4	3	7
b	11	3	0	3
c	5	2	0	2
d	10	3	1	4
f	6	2	2	4
t_0	4	1	1	2
t_1	8	4	1	5
t_2	3	1	1	2
t_3	9	3	1	4
t_4	3	1	1	2
t_5	3	1	1	2
t_6	4	1	1	2
t_7	3	1	1	2
t_8	1	1	1	2



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Chaitin-Briggs Allocator

k-colouring using Chaitin's method

1. Simplify(*a*)

Remove nodes in an arbitrary
order s.t. for node *n*, $D(n) < k$

Push them on a stack

k-colouring using Briggs' method

Since $D(n) < k$, we are
guaranteed to find a
colour for *n*



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Chaitin-Briggs Allocator

k-colouring using Chaitin's method

1. Simplify(a)

Remove nodes in an arbitrary
order s.t. for node n , $D(n) < k$

Push them on a stack

2. Simplify(b)

If the graph is not empty, find
the node with the least spill
cost, spill it, and go back to
step Simplify(a)

k-colouring using Briggs' method



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Push them on a stack

2. Simplify(b)

If the graph is not empty, find
the node with the least spill
cost, spill it, and go back to
step Simplify(a)

3. Colour.

Repeatedly pop the node from
top of the stack, plug it in the
graph and give it a colour
distinct from its neighbours

k-colouring using Briggs' method



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Push them on a stack

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3. Colour.

Repeatedly pop the node from top of the stack, plug it in the graph and give it a colour distinct from its neighbours

k -colouring using Briggs' method

1. Simplify(a)

Remove nodes in an arbitrary order s.t. for node n , $D(n) < k$

Push them on a stack

2. Simplify(b)

If the graph is not empty, mark the nodes as potentially spillable and stack them

If $D(n) \geq k$, we may still find a colour for n if two neighbours of n do not interfere with each other and hence can get the same colour



Chaitin-Briggs Allocator

k-colouring using Chaitin's method

1. Simplify(a)

Remove nodes in an arbitrary order s.t. for node n , $D(n) < k$

Push them on a stack

2. Simplify(b)

If the graph is not k -colourable, mark the node with the highest degree as the node with the highest cost, spill it, and go back to step Simplify(a)

3. Colour.

Repeatedly pop the node from top of the stack, plug it in the graph and give it a colour distinct from its neighbours

k-colouring using Briggs' method

1. Simplify(a)

Remove nodes in an arbitrary order s.t. for node n , $D(n) < k$

Push them on a stack

- Chaitin's wisdom: Make the graph k -colourable, mark the node with the highest degree as the node with the highest cost, spill it, and go back to step Simplify(a)
- Briggs' wisdom: Colour the graph to find out if it is colourable

Repeatedly pop the node from top of the stack, plug it in the graph and give it a colour distinct from its neighbours
If a node cannot be coloured, spill it and go back to step Simplify(a)



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Chaitin-Briggs Allocator

k-colouring using Chaitin's method

1. Simplify(a)

Remove nodes in an arbitrary order s.t. for node n , $D(n) < k$

Push them on a stack

2. Simplify(b)

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3. Colour.

Repeatedly pop the node from top of the stack, plug it in the graph and give it a colour distinct from its neighbours

k-colouring using Briggs' method

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Remove nodes in an arbitrary order s.t. for node n , $D(n) < k$

Push them on a stack

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If the graph is not empty, mark the nodes as potentially spillable and stack them

3. Colour.

Repeatedly pop the node from top of the stack, plug it in the graph and give it a colour distinct from its neighbours
If a node cannot be coloured, spill it and go back to step Simplify(a)



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Spilling Decisions

- Spill cost is weighted by loop nesting depth d

$$C(n) = (L(n) + S(n)) \times 10^d$$

- Sometime people normalize $C(n)$ by degree and consider the ratio $\frac{C(n)}{D(n)}$
- Spill decision should be taken for one live range at a time
 - When we conclude that we need spilling, we should spill a live range with the least cost and restart simplification after spilling the live range
 - in **Simply(b)** step in Chaitin's method
 - in **Colour** step in Briggs' modification
 - Spilling reduces the degree of other live ranges and another round of simplification may give us a better order of colouring the nodes



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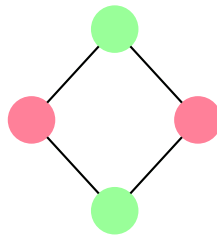
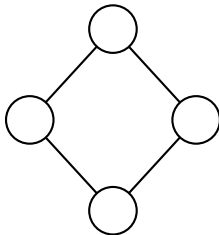
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The Advantage of Delaying Spilling from Simplification to Colouring

- Chaitin's method cannot colour the diamond graph with two colours but Brigg's method can

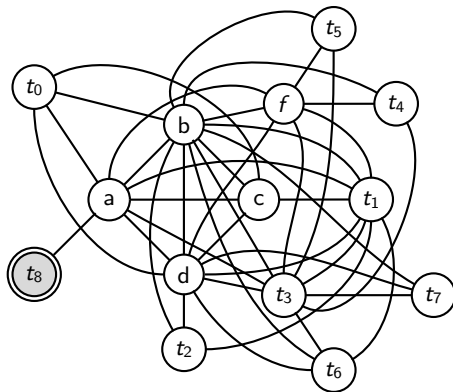


- Chaitin's method would spill live ranges because the degree of live ranges is not smaller than the number of colours
- Brigg's method would not spill before coloring and would find that the two neighbours of any node in this graph can be given the same colour



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



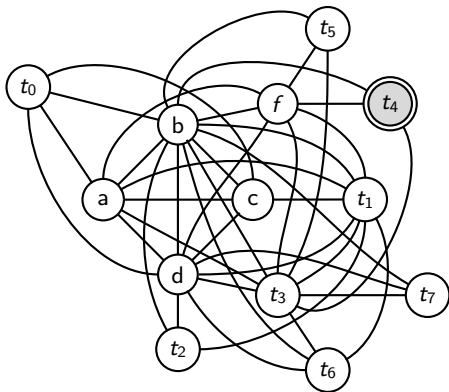
n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	

Step Simplify(a)



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



Step Simplify(a)

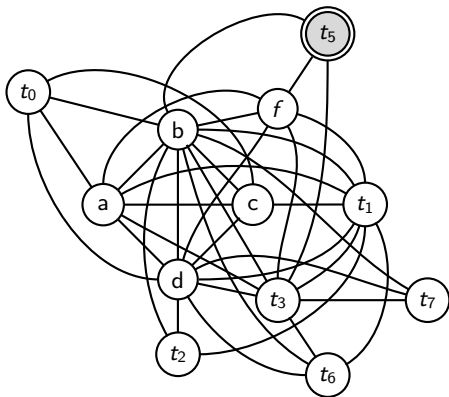
t_8

n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



Step Simplify(a)

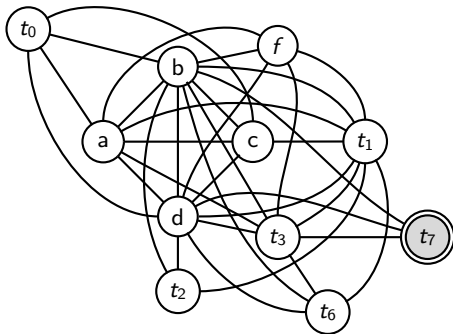
t_4
t_8

n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



Step Simplify(a)

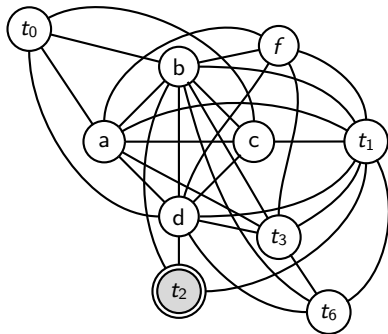
t_5
 t_4
 t_8

n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



Step Simplify(a)

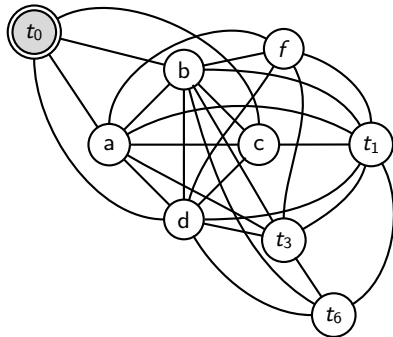
t_7
 t_5
 t_4
 t_8

n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	
c	≥ 5
a	
t_1	
t_3	
f	
d	
b	



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	

t_2
t_7
t_5
t_4
t_8

Step Simplify(a)

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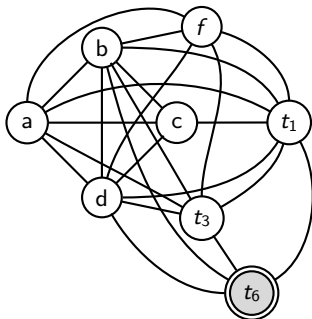
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Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



Step Simplify(a)

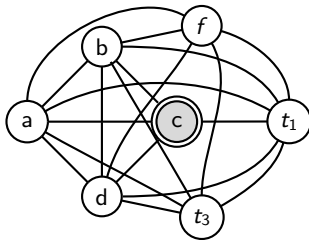
n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	

t_0
 t_2
 t_7
 t_5
 t_4
 t_8



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



t_6
 t_0
 t_2
 t_7
 t_5
 t_4
 t_8

n	$D(n)$
t_8 t_4 t_5 t_7 t_2 t_0 t_6	< 5
c a t_1 t_3 f d b	≥ 5

Step Simplify(a)

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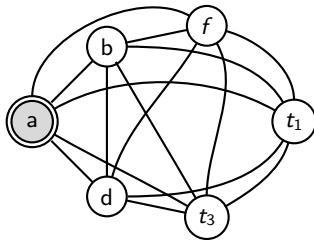
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Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



PS

c
t_6
t_0
t_2
t_7
t_5
t_4
t_8

n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	

Step Simplify(b)



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours

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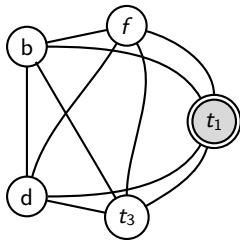
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scip

Instruction Selection



PS	<i>a</i>
PS	<i>c</i>
	<i>t₆</i>
	<i>t₀</i>
	<i>t₂</i>
	<i>t₇</i>
	<i>t₅</i>
	<i>t₄</i>
	<i>t₈</i>

<i>n</i>	$D(n)$
<i>t₈</i>	< 5
<i>t₄</i>	
<i>t₅</i>	
<i>t₇</i>	
<i>t₂</i>	
<i>t₀</i>	
<i>t₆</i>	≥ 5
<i>c</i>	
<i>a</i>	
<i>t₁</i>	
<i>t₃</i>	
<i>f</i>	
<i>d</i>	
<i>b</i>	

Step Simplify(b)



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours

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Topic:

Code Generation

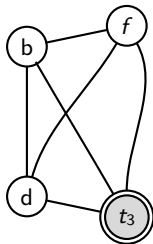
Section:

Global Register
Allocation

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scip

Instruction Selection



PS	t_1
PS	a
PS	c
	t_6
	t_0
	t_2
	t_7
	t_5
	t_4
	t_8

n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	

Step Simplify(b)

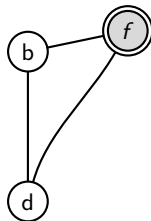


Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours

n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	

PS	t_3
PS	t_1
PS	a
PS	c
	t_6
	t_0
	t_2
	t_7
	t_5
	t_4
	t_8



Step Simplify(b)

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Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours

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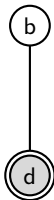
Section:

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Step Simplify(b)

PS	<i>f</i>
PS	<i>t</i> ₃
PS	<i>t</i> ₁
PS	<i>a</i>
PS	<i>c</i>
	<i>t</i> ₆
	<i>t</i> ₀
	<i>t</i> ₂
	<i>t</i> ₇
	<i>t</i> ₅
	<i>t</i> ₄
	<i>t</i> ₈

<i>n</i>	<i>D(n)</i>
<i>t</i> ₈	< 5
<i>t</i> ₄	
<i>t</i> ₅	
<i>t</i> ₇	
<i>t</i> ₂	
<i>t</i> ₀	
<i>t</i> ₆	≥ 5
<i>c</i>	
<i>a</i>	
<i>t</i> ₁	
<i>t</i> ₃	
<i>f</i>	
<i>d</i>	
<i>b</i>	



Simplify and Colour the Interference Graph (Briggs' Method)

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Instruction Selection



PS	d
PS	f
PS	t_3
PS	t_1
PS	a
PS	c
	t_6
	t_0
	t_2
	t_7
	t_5
	t_4
	t_8

n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	≥ 5
c	
a	
t_1	
t_3	
f	
d	
b	

Step Simplify(b)



Simplify and Colour the Interference Graph (Briggs' Method)

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Instruction Selection

PS
PS
PS
PS
PS
PS
PS

b
 d
 f
 t_3
 t_1
 a
 c
 t_6
 t_0
 t_2
 t_7
 t_5
 t_4
 t_8

n	$D(n)$
t_8	< 5
t_4	
t_5	
t_7	
t_2	
t_0	
t_6	
c	≥ 5
a	
t_1	
t_3	
f	
d	
b	



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     

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Instruction Selection

(b)

PS	b					
PS	d					
PS	f					
PS	t_3					
PS	t_1					
PS	a					
PS	c					
	t_6					
	t_0					
	t_2					
	t_7					
	t_5					
	t_4					
	t_8					

Step Colour



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     

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Topic:

Code Generation

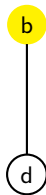
Section:

Global Register
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scip

Instruction Selection



Step Colour

PS	<i>d</i>				
PS	<i>f</i>				
PS	<i>t₃</i>				
PS	<i>t₁</i>				
PS	<i>a</i>				
PS	<i>c</i>				
	<i>t₆</i>				
	<i>t₀</i>				
	<i>t₂</i>				
	<i>t₇</i>				
	<i>t₅</i>				
	<i>t₄</i>				
	<i>t₈</i>				



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     

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Code Generation

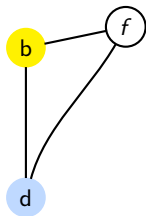
Section:

Global Register
Allocation

Managing Registers
Across Calls

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scip

Instruction Selection



Step Colour

PS	<i>f</i>			
PS	<i>t₃</i>			
PS	<i>t₁</i>			
PS	<i>a</i>			
PS	<i>c</i>			
	<i>t₆</i>			
	<i>t₀</i>			
	<i>t₂</i>			
	<i>t₇</i>			
	<i>t₅</i>			
	<i>t₄</i>			
	<i>t₈</i>			



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     

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Topic:

Code Generation

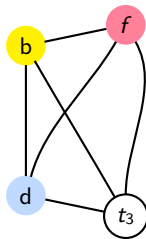
Section:

Global Register
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scip

Instruction Selection



PS	t_3		
PS	t_1		
PS	a		
PS	c		
	t_6		
	t_0		
	t_2		
	t_7		
	t_5		
	t_4		
	t_8		

Step Colour



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     

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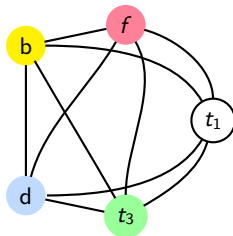
Section:

Global Register
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scip

Instruction Selection



PS	t_1	
PS	a	
PS	c	
	t_6	
	t_0	
	t_2	
	t_7	
	t_5	
	t_4	
	t_8	

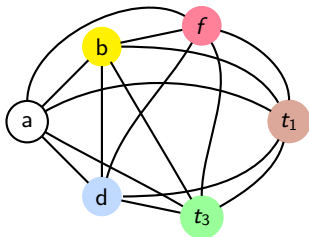
Step Colour

Subgraph K_5 needs all 5 colours



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours



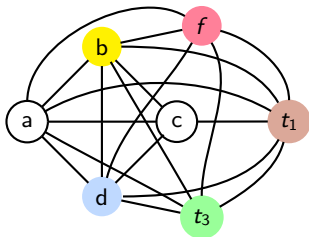
PS	a	No color (Subgraph K_6)
PS	c	Need to spill a and restart simplification
	t_6	
	t_0	Is not required for this example because the degree of c becomes 5
	t_2	
	t_7	
	t_5	
	t_4	
	t_8	

Step Colour



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     



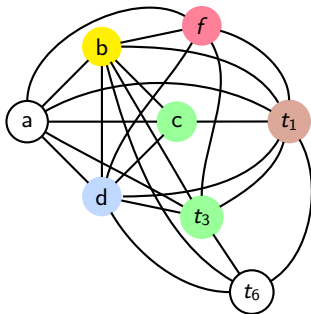
PS	c		
	t ₆		
	t ₀		
	t ₂		
	t ₇		
	t ₅		
	t ₄		
	t ₈		

Step Colour



Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     



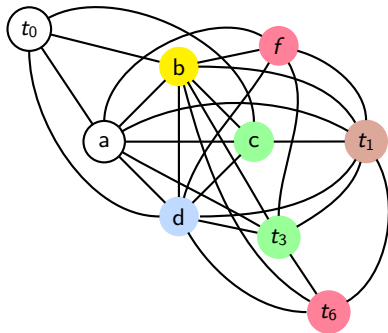
t_6	
t_0	
t_2	
t_7	
t_5	
t_4	
t_8	

Step Colour

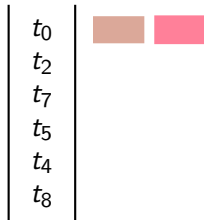


Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     



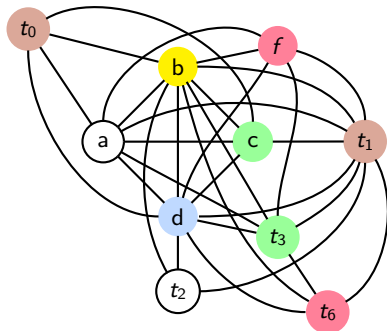
Step Colour





Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     



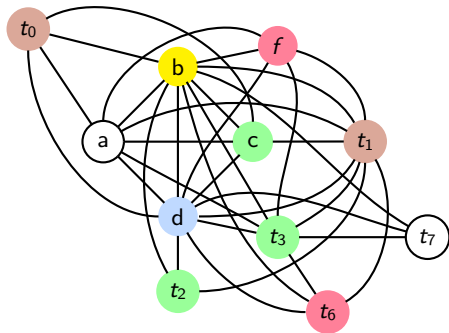
Step Colour



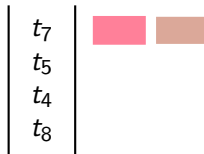


Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     



Step Colour





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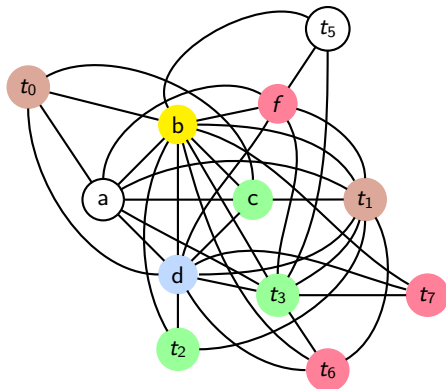
Managing Registers
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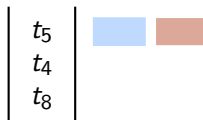
Instruction Selection

Simplify and Colour the Interference Graph (Briggs' Method)

5 Colours     



Step Colour





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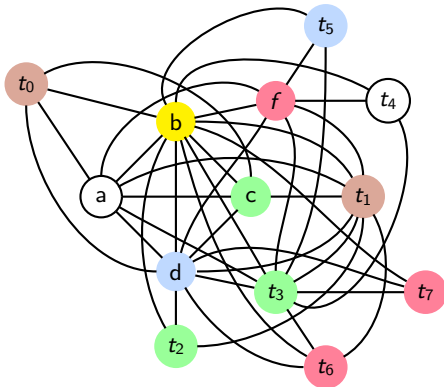
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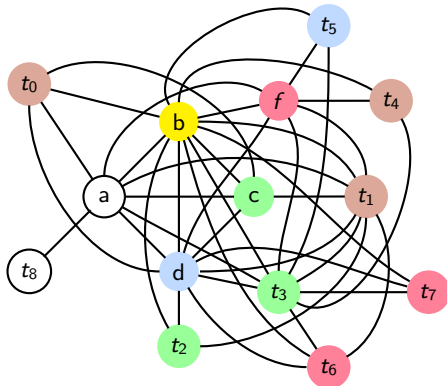
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Simplify and Colour the Interference Graph (Briggs' Method)

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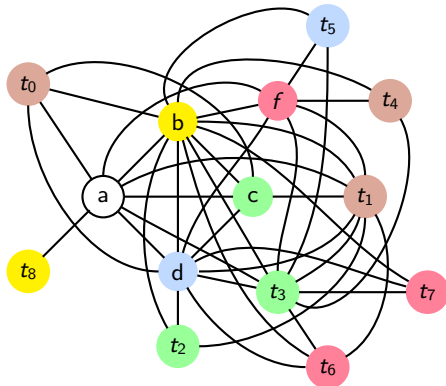
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Simplify and Colour the Interference Graph (Briggs' Method)

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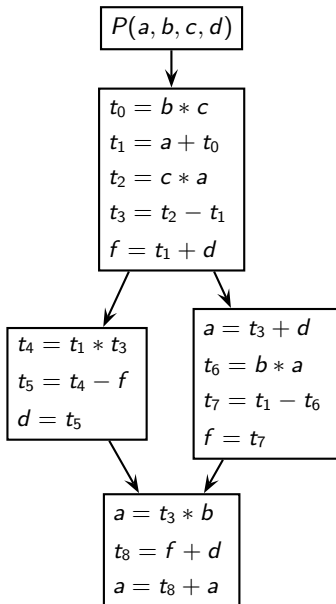
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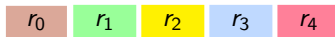
Instruction Selection

Program After Global Register Allocation





Program After Global Register Allocation



$P(a, b, c, d)$

$t_0 = b * c$
 $t_1 = a + t_0$
 $t_2 = c * a$
 $t_3 = t_2 - t_1$
 $f = t_1 + d$

$t_4 = t_1 * t_3$
 $t_5 = t_4 - f$
 $d = t_5$

$a = t_3 + d$
 $t_6 = b * a$
 $t_7 = t_1 - t_6$
 $f = t_7$

$a = t_3 * b$
 $t_8 = f + d$
 $a = t_8 + a$

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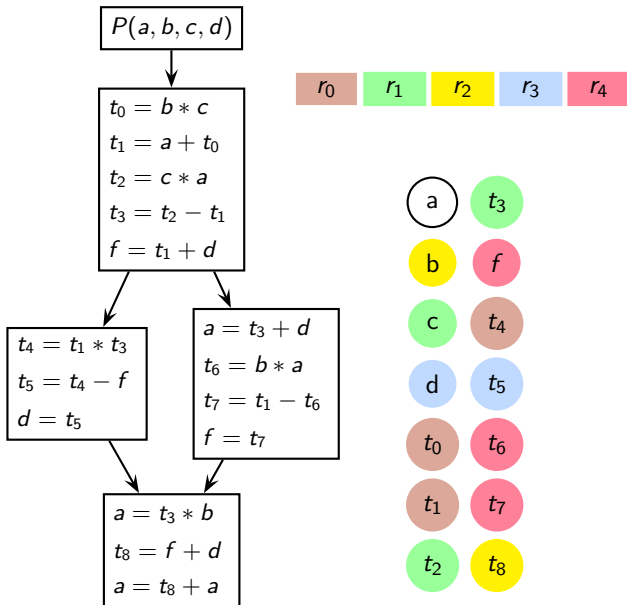
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Program After Global Register Allocation





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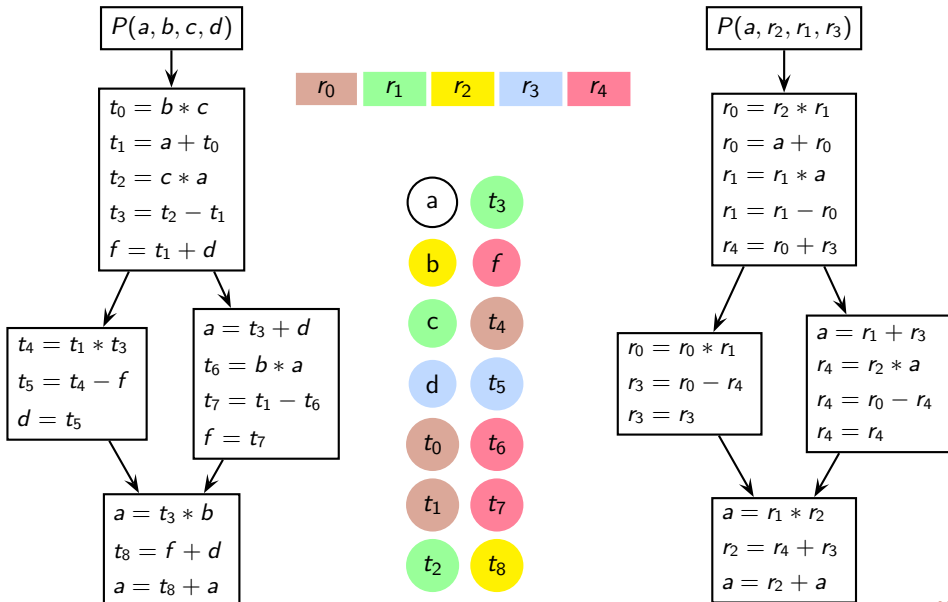
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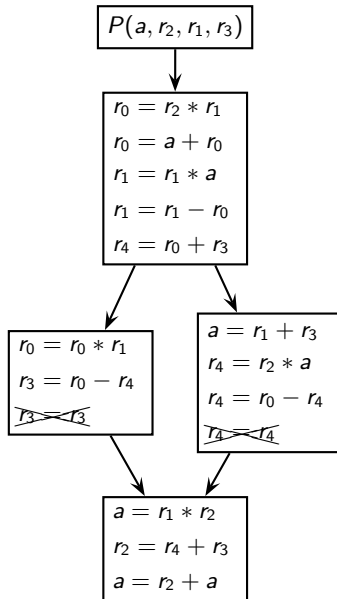
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Instruction Selection

Program After Peephole Optimization





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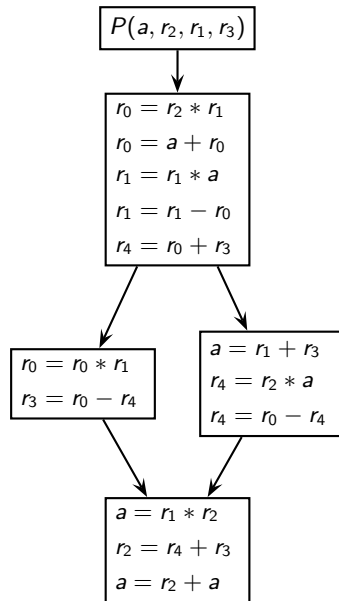
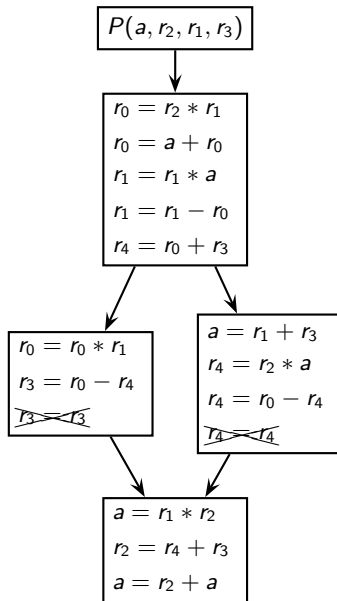
Global Register
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Program After Peephole Optimization





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Live Range Spilling

- Spilling a live range l involves keeping the variable of l in the memory,
 - For RISC architectures: load in a register for every read, store back in the memory for every write
 - For CISC architectures: access directly from memory



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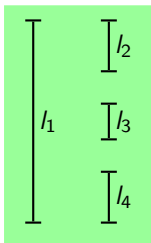
Managing Registers
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Instruction Selection

Live Range Spilling

- Spilling a live range l involves keeping the variable of l in the memory,
 - For RISC architectures: load in a register for every read, store back in the memory for every write
 - For CISC architectures: access directly from memory
- Spilling is necessary if the number of interfering live ranges at a program point exceeds the number of registers



- The degree of l_1 is 3 but at no point does it exceed 2
- 2 registers are sufficient without needing spilling



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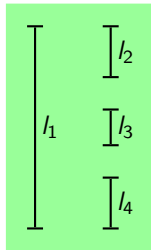
Live Range Splitting

- Splitting a live range l involves creating smaller live ranges $l_1, \dots, l_k$ such that $D(l_i) \leq D(l)$, $1 \leq i \leq k$
 - Live ranges l_i participate in graph colouring and may get a colour if $D(l_i) < D(l)$



Live Range Splitting

- Splitting a live range l involves creating smaller live ranges $l_1, \dots, l_k$ such that $D(l_i) \leq D(l)$, $1 \leq i \leq k$
 - Live ranges l_i participate in graph colouring and may get a colour if $D(l_i) < D(l)$
- Splitting l is useful when it *contains* some live range l' completely and at some point $D(l')$ is smaller than the number of registers



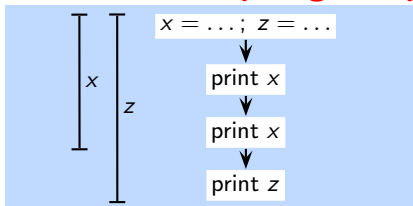
Splitting cannot help us colour this program with a single colour



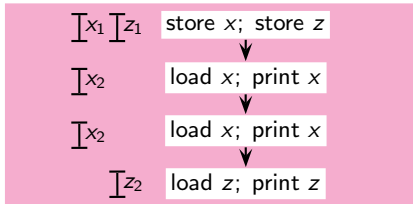
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Instruction Selection

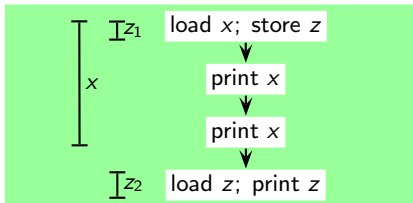
Spilling Vs Splitting



Original program



Splitting x and z leads
to 3 loads and 2 stores



Splitting only z leads
to 2 loads and 1 store

No difference between
splitting or spilling z



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Registers Usage in
sclp

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**Managing Registers
Across Calls**

Registers Usage in
scip

Instruction Selection

Managing Registers Across Calls



Managing Registers Across Calls

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Code Generation

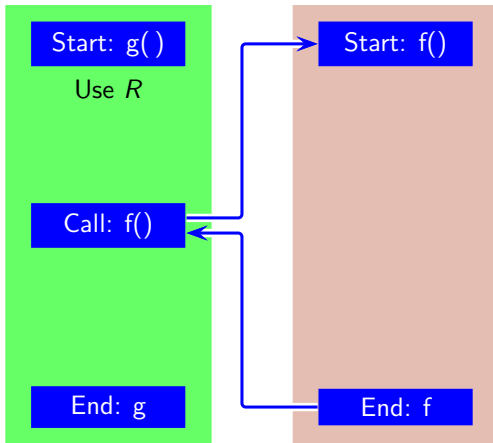
Section:

Global Register
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Instruction Selection



Two options to manage R across the call

- Procedure g saves it before the call and restores it after the call
- Procedure f saves it at the start and restores it the end



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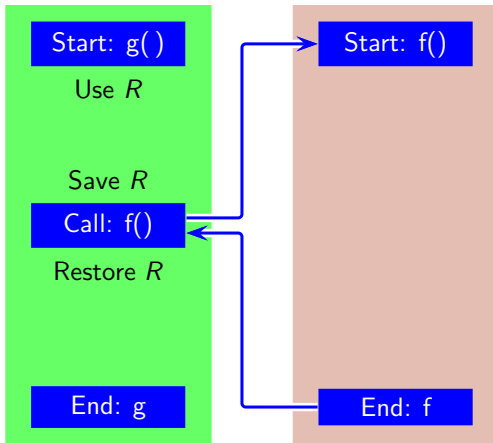
Global Register
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scip

Instruction Selection

Managing Registers Across Calls



If **procedure g** saves **R** before the call and restores it after the call

- It does not know if procedure f really needs R
- However, it knows if the value in R is needed across the call



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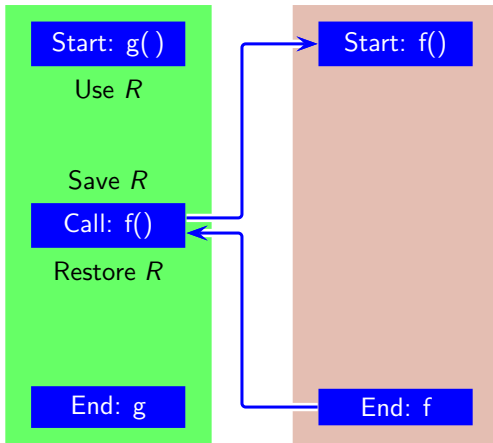
Global Register
Allocation

Managing Registers
Across Calls

Registers Usage in
scip

Instruction Selection

Managing Registers Across Calls



If **procedure g saves R** before the call and restores it after the call

- It does not know if procedure f really needs R
- However, it knows if the value in R is needed across the call

Save and restore would be wasteful if

- f does not need R , or
- the value in R is not needed across the call



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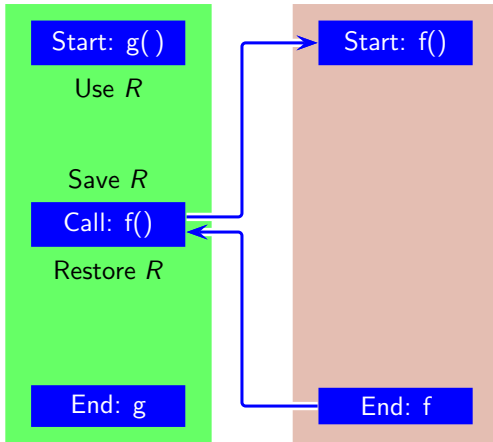
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If **procedure g saves R** before the call and restores it after the call

- It does not know if procedure f really needs R
- However, it knows if the value in R is needed across the call

Save and restore would be wasteful if

- f does not need R , or

Unavoidable

- the value in R is not needed across the call

Avoidable



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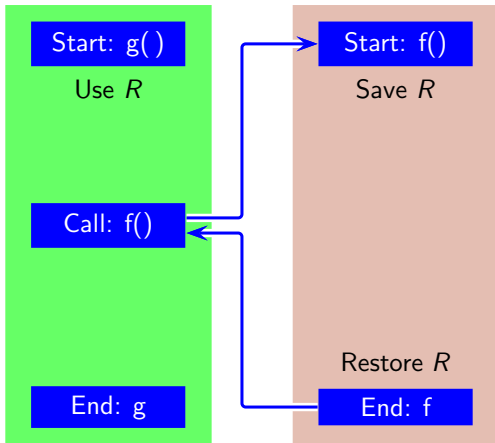
Global Register
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Instruction Selection

Managing Registers Across Calls



If **procedure f saves R** at the start and restores it at the end

- It does not know if procedure g contains a value needed across the call
- However, it knows if R is needed within f



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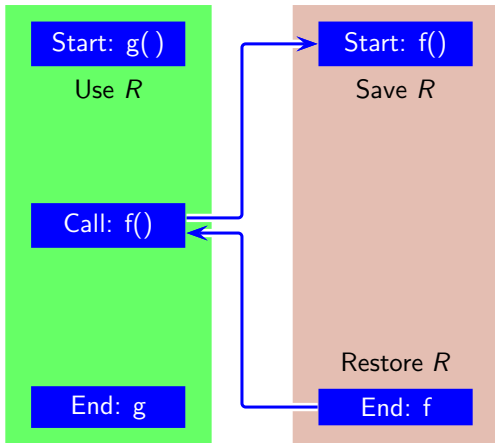
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If **procedure f saves R** at the start and restores it at the end

- It does not know if procedure g contains a value needed across the call
- However, it knows if R is needed within f

Save and restore would be wasteful if

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- f does not need R



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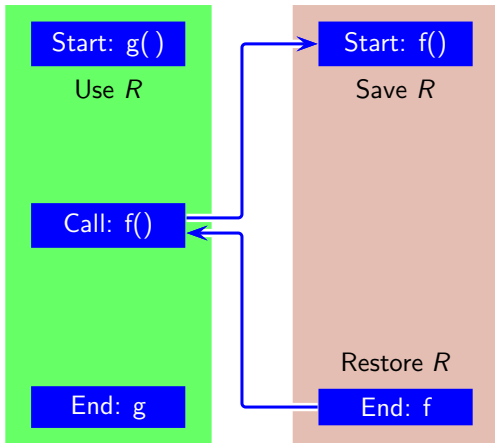
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Instruction Selection

Managing Registers Across Calls



If **procedure f** saves **R** at the start and restores it at the end

- It does not know if procedure g contains a value needed across the call
- However, it knows if R is needed within f

Save and restore would be wasteful if

- the value in R is not needed across the call, or **Unavoidable**
- f does not need R **Available**



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Caller-Saved and Callee-Saved Registers

- Caller-saved register.
 - Saving is at the discretion of the caller
 - Callee can use it without the fear of overwriting useful data
 - Also known as **call-clobbered** register
- Callee-saved register.
 - Saving is at the discretion of the callee
 - Caller can use it without the fear of overwriting useful data
 - Also known as **call-preserved** register



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 - Saving is at the discretion of the callee
 - Caller can use it without the fear of overwriting useful data
 - Also known as **call-preserved** register
- How to use these registers in a procedure?



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 - Also known as **call-preserved** register
- How to use these registers in a procedure?
 - Use a caller-saved register R for values that are not live across a call
 R is not saved by the callee (convention)
nor by the caller because the value is not needed



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 - Also known as **call-preserved** register
- How to use these registers in a procedure?
 - Use a caller-saved register R for values that are not live across a call
 R is not saved by the callee (convention)
nor by the caller because the value is not needed
 - Use a callee-saved register R for values that are live across a call
 R is not saved by the caller (convention)
saved by the callee only if the value is needed



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Caller-Saved and Callee-Saved Registers

- Caller-saved register.

- Saving is at the discretion of the caller
- Callee can use it without the fear of overwriting useful data
- Also known as **call-clobbered** register

- Callee

- S This convention is decided by the architecture (and
- C not by a compiler) to facilitate separate compilation

- A

- How to

- U Different object modules, in particular libraries, may
- F be compiled by different compilers and must follow
- a uniform convention

nor by the caller because the value is not needed

- Use a callee-saved register R for values that are live across a call
- R is not saved by the caller (convention)**
- saved by the callee only if the value is needed**

across a call



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Integrating Caller-Saved and Callee-Saved Registers Within Graph Colouring

- Let **callee-saved** and **caller-saved** registers be denoted by synthetic live ranges coloured **green** and **red** respectively

Start: $f()$

Call: $g()$

End: f



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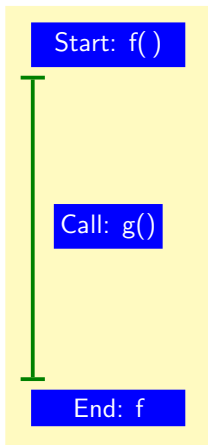
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- Let **callee-saved** and **caller-saved** registers be denoted by synthetic live ranges coloured **green** and **red** respectively
- In f , a **callee-saved** register is assumed to be occupying a value that is live across a call of f in its callers
Construct a **green** live range spanning the entire body of f (indicating that it is not available in the body of f because it is used in the callers of f)



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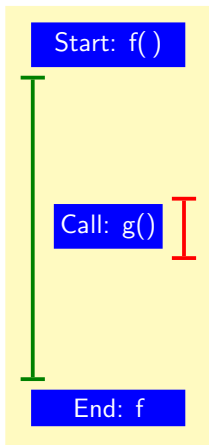
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- In f , a **callee-saved** register is assumed to be occupying a value that is live across a call of f in its callers
Construct a **green** live range spanning the entire body of f (indicating that it is not available in the body of f because it is used in the callers of f)
- In f , a **caller-saved** register is assumed to be occupying a value that is not live across calls within f
Construct a **red** live range spanning the calls in f (indicating that it is not available across the call because it is used in the callees of f)



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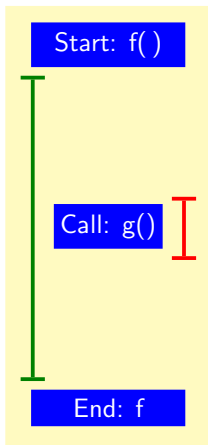
Global Register
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Construct a **green** live range spanning the entire body of f (indicating that it is not available in the body of f because it is used in the callers of f)
- In f , a **caller-saved** register is assumed to be occupying a value that is not live across calls within f
Construct a **red** live range spanning the calls in f (indicating that it is not available across the call because it is used in the callees of f)
- Construct the interference graph with these additional live ranges



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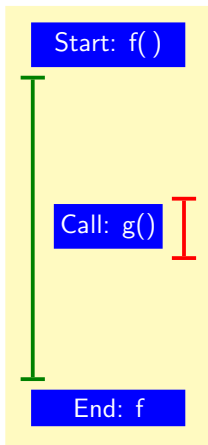
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- In f , a **callee-saved** register is assumed to be occupying a value that is live across a call of f in its callers
Construct a **green** live range spanning the entire body of f (indicating that it is not available in the body of f because it is used in the callers of f)
- In f , a **caller-saved** register is assumed to be occupying a value that is not live across calls within f
Construct a **red** live range spanning the calls in f (indicating that it is not available across the call because it is used in the callees of f)
- Construct the interference graph with these additional live ranges
- Colour the graph with the constraint that the red live ranges cannot be spilled/split.



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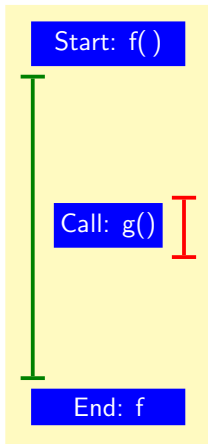
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Instruction Selection

Integrating Caller-Saved and Callee-Saved Registers Within Graph Colouring



- Let **callee-saved** and **caller-saved** registers be denoted by synthetic live ranges coloured **green** and **red** respectively.

- In We may spill/split green or other live ranges (except red) if we cannot colour the graph

- The reason red live range cannot be spilled or split is that it is used in the callee procedure g and f does not have access to the value; in f , we can save a value before the call and restore it but it does not amount to spilling the red live range in the callee g

- The green live range can be spilled by saving the value at the start of f and restoring at the end of f (hence spill cost 2)

- If no live range needs to spill/split, we have avoided all saves and restores across the calls to f and within f

- cannot be spilled/split.



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Example of Register Management Across Calls (2 Registers)

```
void S()  
{  c =...  
   a =...  
   print a  
   print c  
   print a  
   call Q()  
   b =...  
   print b  
   print c  
   print b  
   call T()  
   print c  
}
```

```
void Q()  
{  f =...  
   d =...  
   print d  
   print f  
   print d  
   e =...  
   print e  
   print f  
   print e  
   print f  
}
```



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Example of Register Management Across Calls (2 Registers)

```
void S()  
{  
  c = ...  
  a = ...  
  print a  
  print c  
  print a  
  call Q() r0  
  b = ...  
  print b  
  print c  
  print b  
  call T() r0  
  print c  
}
```

```
void Q()  
{  
  f = ...  
  d = ...  
  print d  
  print f  
  print d  
  e = ...  
  print e  
  print f  
}
```

Identify live ranges in procedure *S*

Add a green live range to represent that a caller of *S* may be freely using the callee-saved register *r1* (which should be saved by *S* in its role as a callee)

Add a red live range across each call to represent that a callee of *S* may be freely using the caller-saved register *r0* (which should be saved by *S* in its role as a caller)



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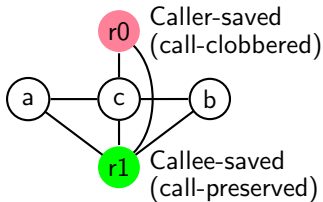
Registers Usage in
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Example of Register Management Across Calls (2 Registers)

```
void S()  
{  
  c = ...  
  a = ...  
  print a  
  print c  
  print a  
  call Q() r0  
  b = ...  
  print b  
  print c  
  print b  
  call T() r0  
  print c  
}
```

```
void Q()  
{  
  f = ...  
  d = ...  
  print d  
  print f  
  print d  
  e = ...  
  print e  
  print f  
  print e  
}
```



Identify live ranges in procedure *S*

Add a green live range to represent that a caller of *S* may be freely using the callee-saved register *r1* (which should be saved by *S* in its role as a callee)

Add a red live range across each call to represent that a callee of *S* may be freely using the caller-saved register *r0* (which should be saved by *S* in its role as a caller)

Construct the interference graph for *S*



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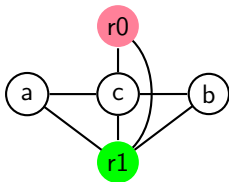
Registers Usage in
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Example of Register Management Across Calls (2 Registers)

```
void S()  
{  c =...  
   a =...  
   print a  
   print c  
   print a  
   call Q()  
   b =...  
   print b  
   print c  
   print b  
   call T()  
   print c  
}
```

```
void Q()  
{  f =...  
   d =...  
   print d  
   print f  
   print d  
   e =...  
   print e  
   print f  
   print e  
   print f  
}
```



Identify live ranges in procedure *S*

Add a green live range to represent that a caller of *S* may be freely using the callee-saved register *r1* (which should be saved by *S* in its role as a callee)

Add a red live range across each call to represent that a callee of *S* may be freely using the caller-saved register *r0* (which should be saved by *S* in its role as a caller)

Construct the interference graph for *S*



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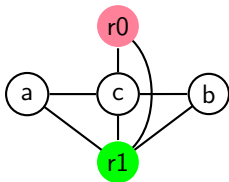
```
void S()  
{ c =...  
  a =...  
  print a  
  print c  
  print a  
  call Q()  
  b =...  
  print b  
  print c  
  print b  
  call T()  
  print c  
}
```

```
void Q()  
{ f =...  
  d =...  
  print d  
  print f  
  print d  
  e =...  
  print e  
  print f  
  print e  
  print f  
}
```

Identify live ranges in procedure Q

Add a green live range to represent that a caller of Q may be freely using the callee-saved register r1 (which should be saved by Q in its role as a callee)

No call in Q so no red live range (or it does not interfere with any live range)





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Example of Register Management Across Calls (2 Registers)

```
void S()  
{ c =...  
  a =...  
  print a  
  print c  
  print a  
  call Q()  
  b =...  
  print b  
  print c  
  print b  
  call T()  
  print c  
}
```

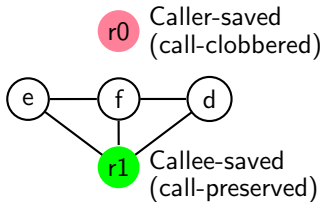
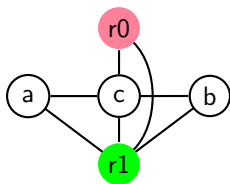
```
void Q()  
{ f =...  
  d =...  
  print d  
  print f  
  print d  
  e =...  
  print e  
  print f  
  print e  
  print f  
}
```

Identify live ranges in procedure Q

Add a green live range to represent that a caller of Q may be freely using the callee-saved register r1 (which should be saved by Q in its role as a callee)

No call in Q so no red live range (or it does not interfere with any live range)

Construct the interference graph for Q





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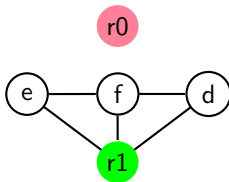
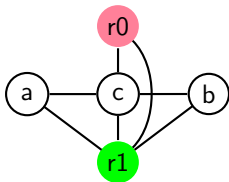
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void S()  
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  c = ...  
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  print a  
  print c  
  print a  
  call Q()  
  b = ...  
  print b  
  print c  
  print b  
  call T()  
  print c  
}
```

```
void Q()  
{  
  f = ...  
  d = ...  
  print d  
  print f  
  print d  
  e = ...  
  print e  
  print f  
  print e  
  print f  
}
```



Identify live ranges in procedure Q

Add a green live range to represent that a caller of Q may be freely using the callee-saved register r1 (which should be saved by Q in its role as a callee)

No call in Q so no red live range (or it does not interfere with any live range)

Construct the interference graph for Q



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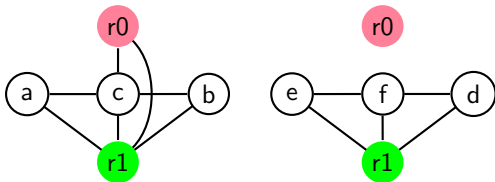
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void S()  
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  print a  
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  b =...  
  print b  
  print c  
  print b  
  call T()  
  print c  
}
```

```
void Q()  
{  f =...  
  d =...  
  print d  
  print f  
  print d  
  e =...  
  print e  
  print f  
  print e  
  print f  
}
```

Cannot colour the interference
graph of *S* with two colours so we
need to spill some live ranges





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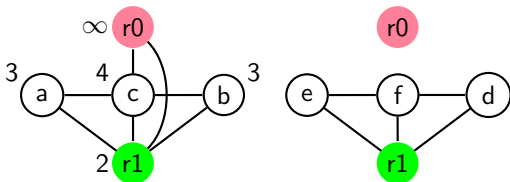
Example of Register Management Across Calls (2 Registers)

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  print c  
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  print c  
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  d = ...  
  print d  
  print f  
  print d  
  e = ...  
  print e  
  print f  
  print e  
  print f  
}
```

Cannot colour the interference
graph of *S* with two colours so we
need to spill some live ranges

Compute the spill costs





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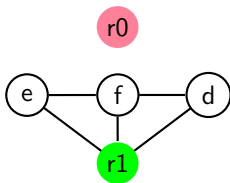
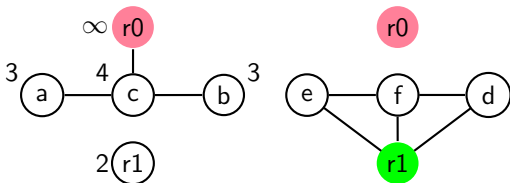
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  print c  
}
```

```
void Q()  
{  
  f = ...  
  d = ...  
  print d  
  print f  
  print d  
  e = ...  
  print e  
  print f  
  print e  
  print f  
}
```

Cannot colour the interference
graph of *S* with two colours so we
need to spill some live ranges

Compute the spill costs

Spill *r1* because it has the least
spill cost





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Example of Register Management Across Calls (2 Registers)

```
void S()  
{  
  c = ...  
  a = ...  
  print a  
  print c  
  print a  
  call Q()  
  b = ...  
  print b  
  print c  
  print b  
  call T()  
  print c  
}
```

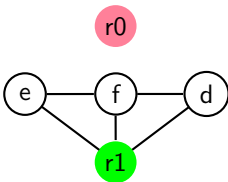
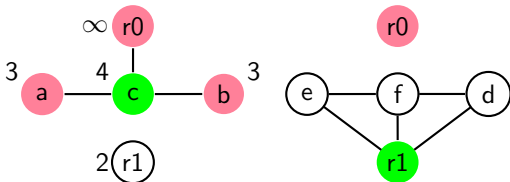
```
void Q()  
{  
  f = ...  
  d = ...  
  print d  
  print f  
  print d  
  e = ...  
  print e  
  print f  
  print e  
  print f  
}
```

Cannot colour the interference
graph of S with two colours so we
need to spill some live ranges

Compute the spill costs

Spill $r1$ because it has the least
spill cost

Give green color to c and red to a
and b





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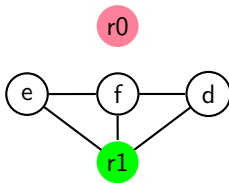
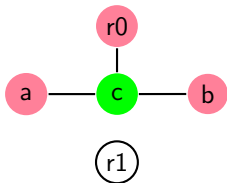
Instruction Selection

Example of Register Management Across Calls (2 Registers)

```
void S()  
{  
  c = ...  
  a = ...  
  print a  
  print c  
  print a  
  call Q()  
  b = ...  
  print b  
  print c  
  print b  
  call T()  
  print c  
}
```

```
void Q()  
{  
  f = ...  
  d = ...  
  print d  
  print f  
  print d  
  e = ...  
  print e  
  print f  
  print e  
  print f  
}
```

Cannot colour the interference
graph of Q with two colours so
we need to spill some live ranges





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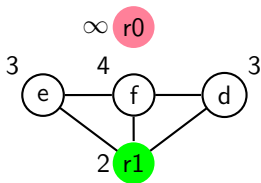
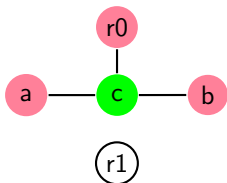
Instruction Selection

Example of Register Management Across Calls (2 Registers)

```
void S()  
{  
  c = ...  
  a = ...  
  print a  
  print c  
  print a  
  call Q()  
  b = ...  
  print b  
  print c  
  print b  
  call T()  
  print c  
}
```

```
void Q()  
{  
  f = ...  
  d = ...  
  print d  
  print f  
  print d  
  e = ...  
  print e  
  print f  
  print e  
  print f  
}
```

Cannot colour the interference
graph of Q with two colours so
we need to spill some live ranges
Compute the spill costs





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Example of Register Management Across Calls (2 Registers)

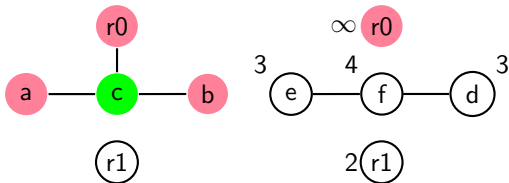
```
void S()  
{  
  c = ...  
  a = ...  
  print a  
  print c  
  print a  
  call Q()  
  b = ...  
  print b  
  print c  
  print b  
  call T()  
  print c  
}
```

```
void Q()  
{  
  f = ...  
  d = ...  
  print d  
  print f  
  print d  
  e = ...  
  print e  
  print f  
  print e  
  print f  
}
```

Cannot colour the interference
graph of Q with two colours so
we need to spill some live ranges

Compute the spill costs

Spill $r1$ because it has the least
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Example of Register Management Across Calls (2 Registers)

```
void S()
{
  c = ...
  a = ...
  print a
  print c
  print a
  call Q()
  b = ...
  print b
  print c
  print b
  call T()
  print c
}
```

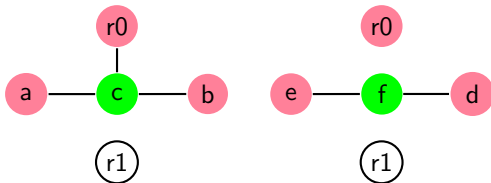
```
void Q()
{
  f = ...
  d = ...
  print d
  print f
  print d
  e = ...
  print e
  print f
  print e
  print f
}
```

Cannot colour the interference graph of Q with two colours so we need to spill some live ranges

Compute the spill costs

Spill $r1$ because it has the least spill cost

Give one color to f and the other color to e and d





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**Registers Usage in
sclp**

Instruction Selection

Registers Usage in sclp



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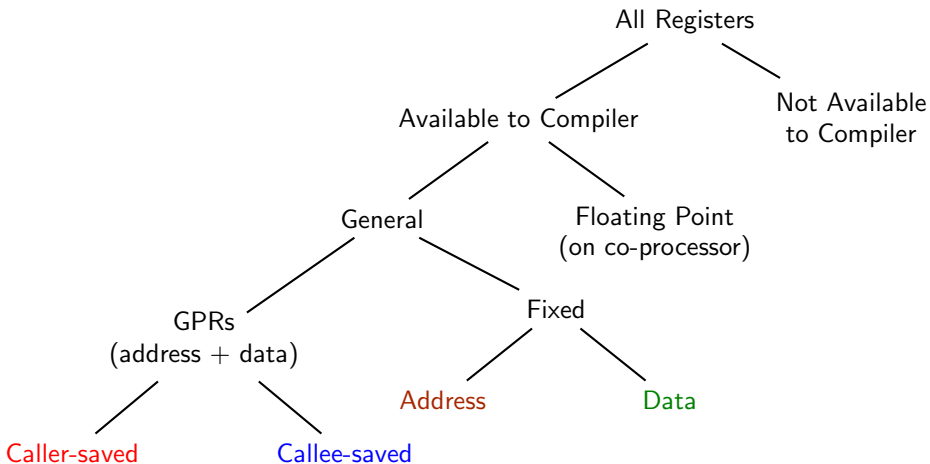
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Register Categories for Spim





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Registers in Spim

Name	No.	Sizes	Use	Category
\$zero	00	32		constant data
at	01	32		NA
v0	02	32,64	expr result	caller-saved
v1	03	32	function result	caller-saved
a0	04	32,64	argument	caller-saved
a1	05	32	argument	caller-saved
a2	06	32,64	argument	caller-saved
a3	07	32	argument	caller-saved
t0	08	32,64	temporary	caller-saved
t1	09	32	temporary	caller-saved
t2	10	32,64	temporary	caller-saved
t3	11	32	temporary	caller-saved
t4	12	32,64	temporary	caller-saved
t5	13	32	temporary	caller-saved
t6	14	32,64	temporary	caller-saved
t7	15	32	temporary	caller-saved

Name	No.	Sizes	Use	Category
s0	16	32,64	temporary	callee-saved
s1	17	32	temporary	callee-saved
s2	18	32,64	result	callee-saved
s3	19	32	result	callee-saved
s4	20	32,64	temporary	callee-saved
s5	21	32	temporary	callee-saved
s6	22	32,64	temporary	callee-saved
s7	23	32	temporary	callee-saved
t8	24	32,64	temporary	caller-saved
t9	25	32	temporary	caller-saved
k0	26	32,64		NA
k1	27	32		NA
gp	28	32,64	global pointer	address
sp	29	32	stack pointer	address
fp	30	32,64	frame pointer	address
ra	31	32	return address	address



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Co-Processor Registers in Spim

Name	Number	Sizes
f0	00	32,64
f1	01	32
f2	02	32,64
f3	03	32
f4	04	32,64
f5	05	32
f6	06	32,64
f7	07	32
f8	08	32,64
f9	09	32
f10	10	32,64
f11	11	32
f12	12	32,64
f13	13	32
f14	14	32,64
f15	15	32

Name	Number	Sizes
f16	16	32,64
f17	17	32
f19	18	32,64
f19	19	32
f20	20	32,64
f21	21	32
f22	22	32,64
f23	23	32
f24	24	32,64
f25	25	32
f26	26	32,64
f27	27	32
f28	28	32,64
f29	29	32
f30	30	32,64
f31	31	32



Registers Used By SCLP for Storing Intermediate Results

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Register		Type
Name	Pair	
v0		int
v1		int
t0		int
t1		int
t2		int
t3		int
t4		int
t5		int
t6		int
t7		int
t8		int
t9		int

Register		Type
Name	Pair	
s0		int
s1		int
s2		int
s3		int
s4		int
s5		int
s6		int
s7		int
f0	f0,f1	float
f2	f2,f3	float
f4	f4,f5	float
f6	f5,f6	float

Register		Type
Name	Pair	
f8	f8,f9	float
f10	f10,f11	float
f12	f12,f13	float
f14	f14,f15	float
f16	f16,f17	float
f18	f18,f19	float
f20	f20,f21	float
f22	f22,f23	float
f24	f24,f25	float
f26	f26,f27	float
f28	f28,f29	float
f30	f30,f31	float



Registers Used By SCLP for Storing Intermediate Results

Designated for function result, hence ignored

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Instruction Selection

Register		Type
Name	Pair	
v0		int
v1		int
t0		int
t1		int
t2		int
t3		int
t4		int
t5		int
t6		int
t7		int
t8		int
t9		int

Register		Type
Name	Pair	
s0		int
s1		int
s2		int
s3		int
s4		int
s5		int
s6		int
s7		int
f0	f0,f1	float
f2	f2,f3	float
f4	f4,f5	float
f6	f5,f6	float

Register		Type
Name	Pair	
f8	f8,f9	float
f10	f10,f11	float
f12	f12,f13	float
f14	f14,f15	float
f16	f16,f17	float
f18	f18,f19	float
f20	f20,f21	float
f22	f22,f23	float
f24	f24,f25	float
f26	f26,f27	float
f28	f28,f29	float
f30	f30,f31	float



Registers Used By SCLP for Storing Intermediate Results

Designated for function result, hence ignored

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Instruction Selection

Register		Type
Name	Pair	
v0		int
t0		int
t1		int
t2		int
t3		int
t4		int
t5		int
t6		int
t7		int
t8		int
t9		int

Register		Type
Name	Pair	
s0		int
s1		int
s2		int
s3		int
s4		int
s5		int
s6		int
s7		int
f2	f2,f3	float
f4	f4,f5	float
f6	f5,f6	float

Register		Type
Name	Pair	
f8	f8,f9	float
f10	f10,f11	float
f12	f12,f13	float
f14	f14,f15	float
f16	f16,f17	float
f18	f18,f19	float
f20	f20,f21	float
f22	f22,f23	float
f24	f24,f25	float
f26	f26,f27	float
f28	f28,f29	float
f30	f30,f31	float



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Register Allocation Policy Used by scip

Very simple strategy

- No register is occupied across any assignment in the source program
The result is stored in memory for the LHS variable and the result register is freed
- Within an expression, values of source variables are loaded into registers
- Intermediate values within an expression (stored in a temporary variable) are assigned a register
The result of a ternary expression is a “saved” temporary which is treated like a source variable
- Getting a new register
 - When a temporary is assigned a register, mark it as occupied
 - When a temporary is used in a TAC statement, mark the register as free
 - To get a new register,
 - Traverse the list of registers and assign the first free register
 - Need to match the type
- Registers are chosen for a TAC statement in this order: first operand, result, second operand (because second operand may not exist)



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Asgn_TAC Stmt

Compute_TAC Stmt

Call_TAC Stmt

Goto_TAC Stmt

IfGoto_TAC Stmt

Return_TAC Stmt

Label_TAC Stmt

IO_TAC Stmt

NOP_TAC Stmt

Move_RTL Stmt

Compute_RTL Stmt

Call_RTL Stmt

Goto_RTL Stmt

IfGoto_RTL Stmt

Return_RTL Stmt

Label_RTL Stmt

Read_RTL Stmt

Write_RTL Stmt

NOP_RTL Stmt

Move_ASM Stmt

Compute_ASM Stmt

Call_ASM Stmt

Goto_ASM Stmt

IfGoto_ASM Stmt

Return_ASM Stmt

Label_ASM Stmt

Syscall_ASM Stmt

NOP_ASM Stmt



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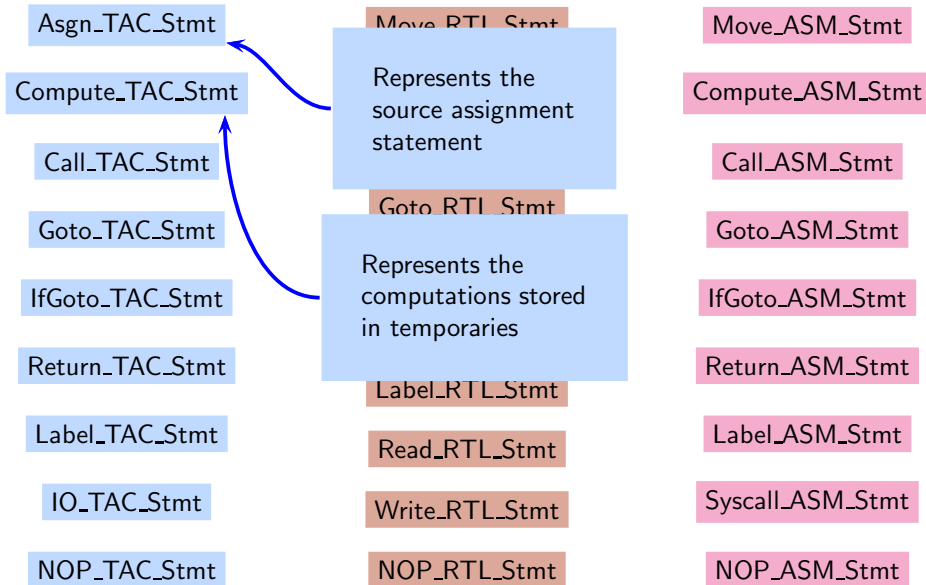
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Instruction Selection in scip

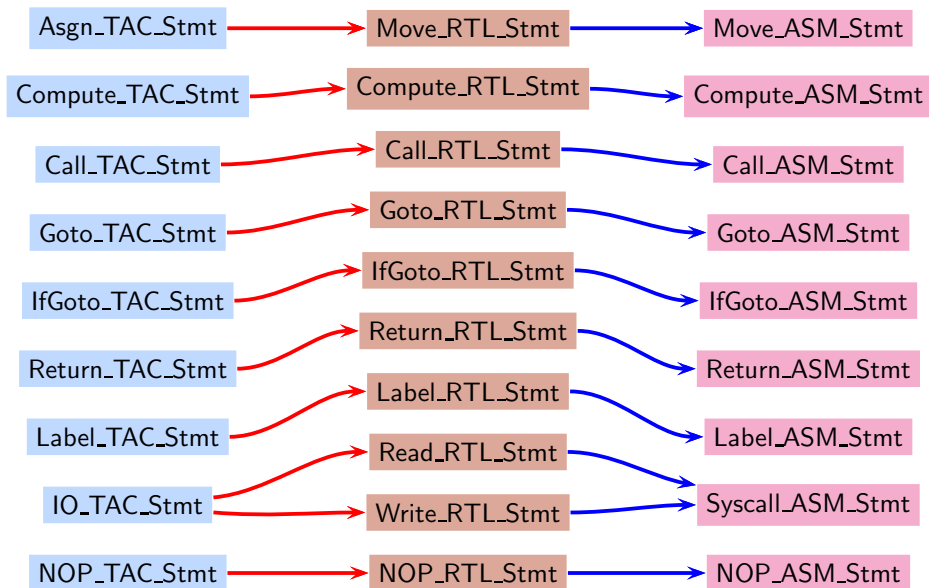




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Instruction Selection in sc1p





Integrated Instruction Selection and Register Allocation Algorithms

For expression trees

- Sethi-Ullman Algorithm
 - Simple machine model (handles RISC architectures well)
 - Optimal in terms of the number of instructions with the minimum number of registers and minimum number of stores
 - Linear in the size of the expression tree
- Aho-Johnson Algorithm
 - Very general machine model (handles CISC architectures also well)
 - Optimal in terms of the cost of execution
 - Linear in the size of the expression tree
(exponential in the arity of instructions which is bounded by a small constant, say 3 or 4)

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Integrated Instruction Selection and Register Allocation Algorithms

For expression trees

- Sethi-Ullman Algorithm
 - Simple machine model (handles RISC architectures well)
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(exponential in the arity of instructions which is bounded by a small constant, say 3 or 4)

For other IR statements, instruction selection is relatively easier and simple methods work well

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Integrated Instruction Selection and Register Allocation Algorithms

For expression trees

- Sethi-Ullman Algorithm

We will cover this

- Simple machine model (handles RISC architectures well)
- Optimal in terms of the number of instructions with the minimum number of registers and minimum number of stores
- Linear in the size of the expression tree

- Aho-Johnson Algorithm

No time for this

- Very general machine model (handles CISC architectures also well)
- Optimal in terms of the cost of execution
- Linear in the size of the expression tree
(exponential in the arity of instructions which is bounded by a small constant, say 3 or 4)

For other IR statements, instruction selection is relatively easier and simple methods work well

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Sethi-Ullman Algorithm: Target Model

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Instruction Selection

- A finite set of registers $r_0, r_1, \dots, r_k$
- Countable memory locations
- Simple machine instructions
 - Load instruction $r \leftarrow m$
 - Store instruction $m \leftarrow r$
 - Compute instructions
 - $r \leftarrow r \text{ op } m$ (result and left operands are same, right operand is in memory)
 - $r_1 \leftarrow r_1 \text{ op } r_2$ (result and left operands are same, right operand is in register)



Sethi-Ullman Algorithm: Input IR

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Instruction Selection

- Expression tree (AST of expressions)
 - no control flow (so no ternary expression)
 - no assignments to source variables (so no side effects),
 - no function calls,
 - no sharing of value (so no DAGs, only trees)
- Algebraic properties such as commutativity and associative not assumed in the basic algorithm
Extended algorithm handles them



Handling a DAG

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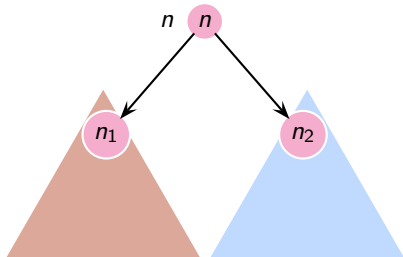
- Generating code to evaluate the shared subexpression only once could enhance efficiency
- Sethi-Ullman algorithm can be made to handle this in the following manner
 - Treat the shared subexpression as a separate expression tree, generate code for it using the Sethi-Ullman algorithm and save the result in a temporary
 - Convert the input DAG into a tree by replacing the subtree by the name of the temporary and replicate it
 - Generate the code using Sethi-Ullman algorithm



The Key Idea Behind Sethi-Ullman Algorithm

The register usage in the code fragment for a tree rooted at n can be described by

- $R(n)$: The number of registers used by the code
- $L(n)$: The number of registers live outside the code for n (i.e., the intermediate results that are required outside the code for n)
- The algorithm minimizes $R(n)$ to avoid storing intermediate results



If the code computes n_1 first, then

$$R(n) = \max(R(n_1), L(n_1) + R(n_2))$$

If the code computes n_2 first, then

$$R(n) = \max(R(n_2), L(n_2) + R(n_1))$$



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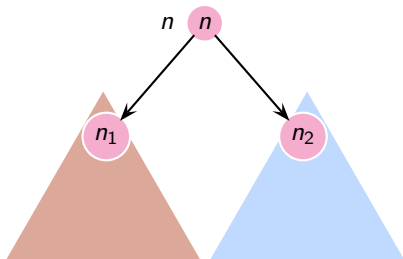
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The Key Idea Behind Sethi-Ullman Algorithm



If the code computes n_1 first, then

$$R(n) = \max(R(n_1), L(n_1) + R(n_2))$$

If the code computes n_2 first, then

$$R(n) = \max(R(n_2), L(n_2) + R(n_1))$$

In order to minimize $R(n)$,

1. Minimize $L(n_1)$ and $L(n_2)$

Minimizing to 0 would introduce a store so minimize to 1

2. Decide whether to evaluate n_1 first or n_2 first



Key Idea #1: Contiguous Evaluation Minimizes $L(n)$ to 1

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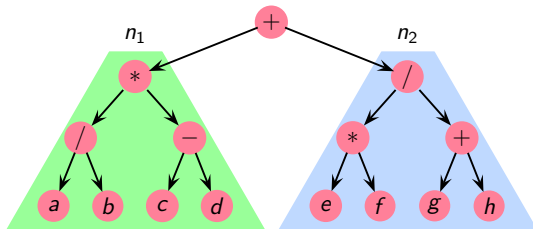
- **Contiguous Evaluation.** Evaluate n_1 completely before evaluating n_2 or vice-versa
 - If we evaluate a subtree completely, we need to hold only the final result in a register during the evaluation of the other subtrees
 - If we do not evaluate a subtree completely before moving to the other subtree, we may have to hold multiple intermediate results in a register during the evaluation of the other subtree
 - This increases the need of registers
- **Strongly Contiguous Evaluation.** All subtrees of n_1 and n_2 are also evaluated contiguously



Key Idea #1: Contiguous Evaluation Minimizes $L(n)$ to 1

Instructions

```
 $r \leftarrow r \text{ op } m$   
 $r_1 \leftarrow r_1 \text{ op } r_2$   
 $r \leftarrow m$   
 $m \leftarrow r$ 
```



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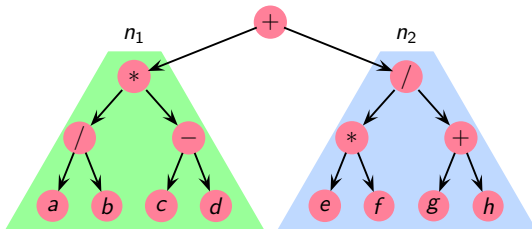
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Instruction Selection



Key Idea #1: Contiguous Evaluation Minimizes $L(n)$ to 1

Instructions

$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$
$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 / b \\ r_1 &\leftarrow c \\ r_1 &\leftarrow r_1 - d \\ r_0 &\leftarrow r_0 * r_1 \end{aligned}$$
$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 / b \\ r_1 &\leftarrow c \\ r_1 &\leftarrow r_1 - d \end{aligned}$$
$$\begin{aligned} r_1 &\leftarrow e \\ r_1 &\leftarrow r_1 * f \\ r_2 &\leftarrow g \\ r_2 &\leftarrow r_2 + h \\ r_1 &\leftarrow r_1 / r_2 \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$
$$\begin{aligned} r_2 &\leftarrow e \\ r_2 &\leftarrow r_2 * f \\ r_3 &\leftarrow g \\ r_3 &\leftarrow r_3 + h \\ r_0 &\leftarrow r_0 * r_1 \\ r_2 &\leftarrow r_2 / r_3 \\ r_0 &\leftarrow r_0 + r_2 \end{aligned}$$


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Key Idea #1: Contiguous Evaluation Minimizes $L(n)$ to 1

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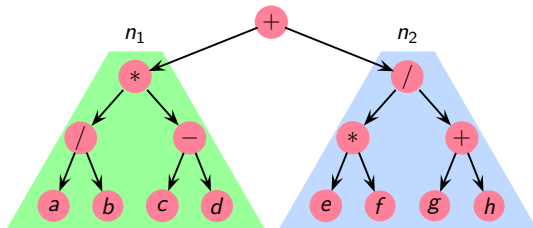
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Instruction Selection

Instructions

```
 $r \leftarrow r \text{ op } m$   
 $r_1 \leftarrow r_1 \text{ op } r_2$   
 $r \leftarrow m$   
 $m \leftarrow r$ 
```



Contiguous

$L(n_1) = 1$

```
 $r_0 \leftarrow a$   
 $r_0 \leftarrow r_0 / b$   
 $r_1 \leftarrow c$   
 $r_1 \leftarrow r_1 - d$   
 $r_0 \leftarrow r_0 * r_1$ 
```

```
 $r_1 \leftarrow e$   
 $r_1 \leftarrow r_1 * f$   
 $r_2 \leftarrow g$   
 $r_2 \leftarrow r_2 + h$   
 $r_1 \leftarrow r_1 / r_2$   
 $r_0 \leftarrow r_0 + r_1$ 
```

```
 $r_0 \leftarrow a$   
 $r_0 \leftarrow r_0 / b$   
 $r_1 \leftarrow c$   
 $r_1 \leftarrow r_1 - d$ 
```

```
 $r_2 \leftarrow e$   
 $r_2 \leftarrow r_2 * f$   
 $r_3 \leftarrow g$   
 $r_3 \leftarrow r_3 + h$ 
```

```
 $r_0 \leftarrow r_0 * r_1$   
 $r_2 \leftarrow r_2 / r_3$   
 $r_0 \leftarrow r_0 + r_2$ 
```



Key Idea #1: Contiguous Evaluation Minimizes $L(n)$ to 1

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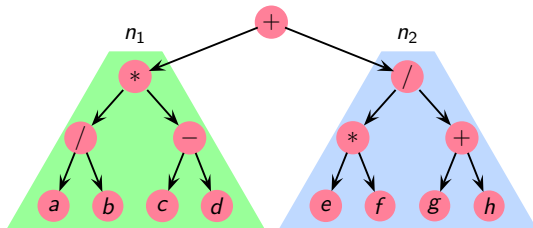
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Instructions

```
 $r \leftarrow r \text{ op } m$   
 $r_1 \leftarrow r_1 \text{ op } r_2$   
 $r \leftarrow m$   
 $m \leftarrow r$ 
```



Contiguous
 $L(n_1) = 1$

```
 $r_0 \leftarrow a$   
 $r_0 \leftarrow r_0 / b$   
 $r_1 \leftarrow c$   
 $r_1 \leftarrow r_1 - d$   
 $r_0 \leftarrow r_0 * r_1$ 
```

```
 $r_1 \leftarrow e$   
 $r_1 \leftarrow r_1 * f$   
 $r_2 \leftarrow g$   
 $r_2 \leftarrow r_2 + h$   
 $r_1 \leftarrow r_1 / r_2$   
 $r_0 \leftarrow r_0 + r_1$ 
```

Non-contiguous
 $L(n_1) = 2$

```
 $r_0 \leftarrow a$   
 $r_0 \leftarrow r_0 / b$   
 $r_1 \leftarrow c$   
 $r_1 \leftarrow r_1 - d$ 
```

```
 $r_2 \leftarrow e$   
 $r_2 \leftarrow r_2 * f$   
 $r_3 \leftarrow g$   
 $r_3 \leftarrow r_3 + h$ 
```

```
 $r_0 \leftarrow r_0 * r_1$   
 $r_2 \leftarrow r_2 / r_3$   
 $r_0 \leftarrow r_0 + r_2$ 
```



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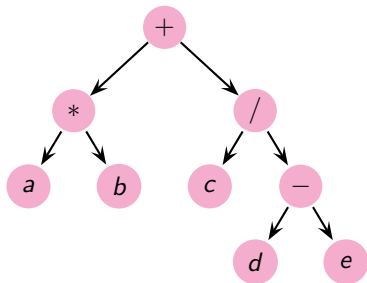
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Key Idea #2: The Minimized $L(n_1)$ Can be “Absorbed” by Changing the Order of Evaluation

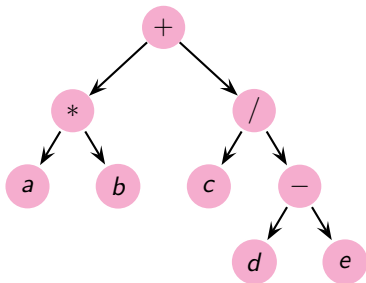
Instructions

$$r \leftarrow r \text{ op } m$$
$$r_1 \leftarrow r_1 \text{ op } r_2$$
$$r \leftarrow m$$
$$m \leftarrow r$$




Key Idea #2: The Minimized $L(n_1)$ Can be “Absorbed” by Changing the Order of Evaluation

Instructions

$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$
$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_1 &\leftarrow c \\ r_2 &\leftarrow d \\ r_2 &\leftarrow r_2 - e \\ r_1 &\leftarrow r_1 / r_2 \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$
$$\begin{aligned} r_1 &\leftarrow c \\ r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow r_1 / r_0 \\ r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$
$$\begin{aligned} r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow c \\ r_1 &\leftarrow r_1 / r_0 \\ r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$


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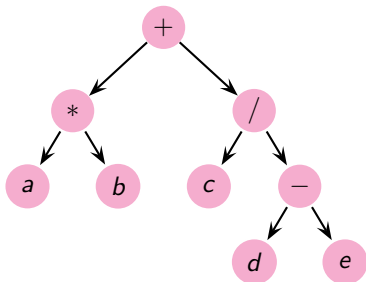
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Key Idea #2: The Minimized $L(n_1)$ Can be “Absorbed” by Changing the Order of Evaluation

Instructions

$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$


Program 1
Regs r_0, r_1, r_2

$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_1 &\leftarrow c \\ r_2 &\leftarrow d \\ r_2 &\leftarrow r_2 - e \\ r_1 &\leftarrow r_1 / r_2 \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$

Program 2
Regs r_0, r_1

$$\begin{aligned} r_1 &\leftarrow c \\ r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow r_1 / r_0 \\ r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$

Program 3
Regs r_0, r_1

$$\begin{aligned} r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow c \\ r_1 &\leftarrow r_1 / r_0 \\ r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$



Key Idea #2: The Minimized $L(n_1)$ Can be “Absorbed” by Changing the Order of Evaluation

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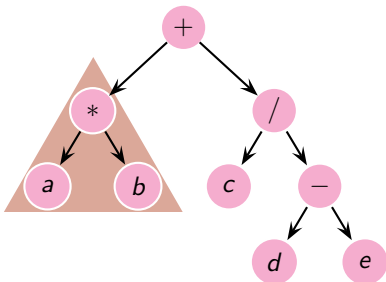
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Instruction Selection

Instructions

$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$


Program 1
Regs r_0, r_1, r_2

$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \end{aligned}$$

$$\begin{aligned} r_1 &\leftarrow c \\ r_2 &\leftarrow d \\ r_2 &\leftarrow r_2 - e \\ r_1 &\leftarrow r_1 / r_2 \end{aligned}$$

$$r_0 \leftarrow r_0 + r_1$$

Program 2
Regs r_0, r_1

$$\begin{aligned} r_1 &\leftarrow c \\ r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow r_1 / r_0 \end{aligned}$$

$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \end{aligned}$$

$$r_0 \leftarrow r_0 + r_1$$

Program 3
Regs r_0, r_1

$$\begin{aligned} r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow c \\ r_1 &\leftarrow r_1 / r_0 \end{aligned}$$

$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \end{aligned}$$

$$r_0 \leftarrow r_0 + r_1$$



Key Idea #2: The Minimized $L(n_1)$ Can be “Absorbed” by Changing the Order of Evaluation

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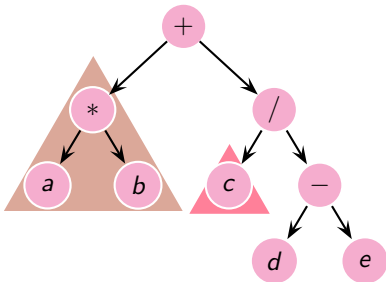
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Instruction Selection

Instructions

$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$


Program 1
Regs r_0, r_1, r_2

$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_1 &\leftarrow c \\ r_2 &\leftarrow d \\ r_2 &\leftarrow r_2 - e \\ r_1 &\leftarrow r_1 / r_2 \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$

Program 2
Regs r_0, r_1

$$\begin{aligned} r_1 &\leftarrow c \\ r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow r_1 / r_0 \\ r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$

Program 3
Regs r_0, r_1

$$\begin{aligned} r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow c \\ r_1 &\leftarrow r_1 / r_0 \\ r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$



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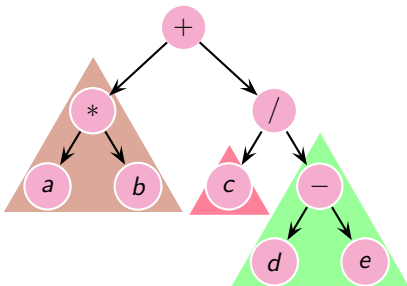
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Key Idea #2: The Minimized $L(n_1)$ Can be “Absorbed” by Changing the Order of Evaluation

Instructions

$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$


Program 1
Regs r_0, r_1, r_2

$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \end{aligned}$$
$$\begin{aligned} r_1 &\leftarrow c \\ r_2 &\leftarrow d \\ r_2 &\leftarrow r_2 - e \end{aligned}$$
$$\begin{aligned} r_1 &\leftarrow r_1 / r_2 \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$

Program 2
Regs r_0, r_1

$$\begin{aligned} r_1 &\leftarrow c \\ r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow r_1 / r_0 \end{aligned}$$
$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$

Program 3
Regs r_0, r_1

$$\begin{aligned} r_0 &\leftarrow d \\ r_0 &\leftarrow r_0 - e \\ r_1 &\leftarrow c \\ r_1 &\leftarrow r_1 / r_0 \end{aligned}$$
$$\begin{aligned} r_0 &\leftarrow a \\ r_0 &\leftarrow r_0 * b \\ r_0 &\leftarrow r_0 + r_1 \end{aligned}$$



Key Idea #2: The Minimized $L(n_1)$ Can be “Absorbed” by Changing the Order of Evaluation

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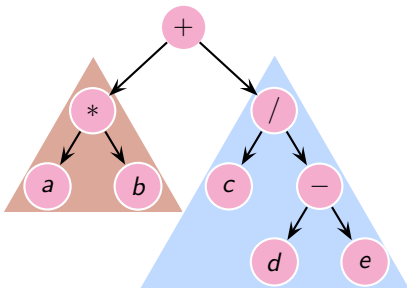
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Instruction Selection

Instructions

```

r ← r op m
r1 ← r1 op r2
r ← m
m ← r
    
```



Program 1
Regs r_0, r_1, r_2

```

r0 ← a
r0 ← r0 * b
    
```

```

r1 ← c
r2 ← d
r2 ← r2 - e
r1 ← r1 / r2
    
```

$r_0 \leftarrow r_0 + r_1$

Program 2
Regs r_0, r_1

```

r1 ← c
r0 ← d
r0 ← r0 - e
r1 ← r1 / r0
    
```

```

r0 ← a
r0 ← r0 * b
    
```

$r_0 \leftarrow r_0 + r_1$

Program 3
Regs r_0, r_1

```

r0 ← d
r0 ← r0 - e
r1 ← c
r1 ← r1 / r0
    
```

```

r0 ← a
r0 ← r0 * b
    
```

$r_0 \leftarrow r_0 + r_1$



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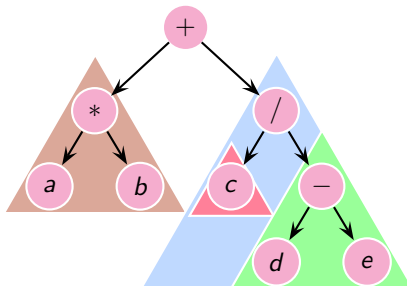
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Instruction Selection

Key Idea #2: The Minimized $L(n_1)$ Can be “Absorbed” by Changing the Order of Evaluation

Instructions

```
 $r \leftarrow r \text{ op } m$   
 $r_1 \leftarrow r_1 \text{ op } r_2$   
 $r \leftarrow m$   
 $m \leftarrow r$ 
```



Program 1
Regs r_0, r_1, r_2

```
 $r_0 \leftarrow a$   
 $r_0 \leftarrow r_0 * b$ 
```

```
 $r_1 \leftarrow c$   
 $r_2 \leftarrow d$   
 $r_2 \leftarrow r_2 - e$   
 $r_1 \leftarrow r_1 / r_2$ 
```

$r_0 \leftarrow r_0 + r_1$

Program 2
Regs r_0, r_1

```
 $r_1 \leftarrow c$   
 $r_0 \leftarrow d$   
 $r_0 \leftarrow r_0 - e$   
 $r_1 \leftarrow r_1 / r_0$ 
```

```
 $r_0 \leftarrow a$   
 $r_0 \leftarrow r_0 * b$ 
```

$r_0 \leftarrow r_0 + r_1$

Program 3
Regs r_0, r_1

```
 $r_0 \leftarrow d$   
 $r_0 \leftarrow r_0 - e$   
 $r_1 \leftarrow c$   
 $r_1 \leftarrow r_1 / r_0$ 
```

```
 $r_0 \leftarrow a$   
 $r_0 \leftarrow r_0 * b$ 
```

$r_0 \leftarrow r_0 + r_1$

The ordering between the pink and green subtrees does not matter (programs 2 and 3)

The ordering between the brown and lightblue subtrees affects the number of registers

We want to minimise the number of registers so that we do not need to store an intermediate result in memory



The Sethi-Ullman Algorithm

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Instruction Selection

- Traverse the expression tree bottom up and label each node with the minimum number of registers needed to evaluate the subexpression rooted at the node
- Traverse the expression tree top down and generate code



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Instruction Selection

- Traverse the expression tree bottom up and label each node with the minimum number of registers needed to evaluate the subexpression rooted at the node
Each node is processed exactly once
- Traverse the expression tree top down and generate code
Each node is processed exactly once



The Sethi-Ullman Algorithm

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Instruction Selection

- Traverse the expression tree bottom up and label each node with the minimum number of registers needed to evaluate the subexpression rooted at the node
Each node is processed exactly once
- Traverse the expression tree top down and generate code
Each node is processed exactly once
- The algorithm is linear in the size of the expression tree



Labelling the Expression Tree

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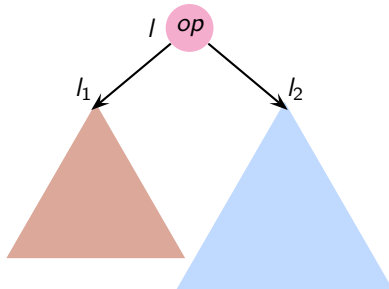
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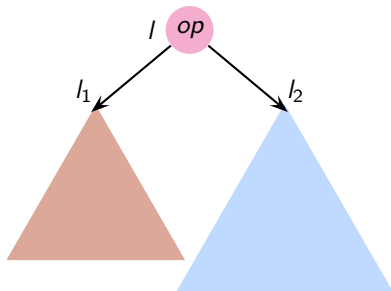
Instruction Selection





Labelling the Expression Tree

Assume that the register requirements of the two subtrees are l_1 and l_2 and that $l_1 < l_2$



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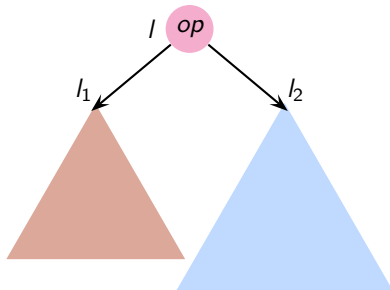
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Instruction Selection

Labelling the Expression Tree

Assume that the register requirements of the two subtrees are l_1 and l_2 and that $l_1 < l_2$

- If we evaluate the brown subtree first, we need l_1 registers to evaluate it, 1 register to hold its result and l_2 registers to evaluate the blue subtree



$$l = \max(l_1, l_2 + 1) = l_2 + 1$$



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Labelling the Expression Tree

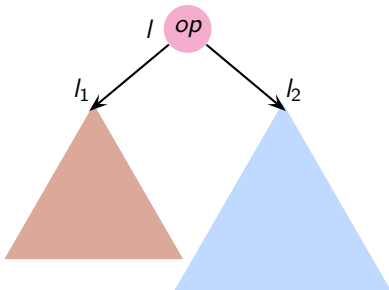
Assume that the register requirements of the two subtrees are l_1 and l_2 and that $l_1 < l_2$

- If we evaluate the brown subtree first, we need l_1 registers to evaluate it, 1 register to hold its result and l_2 registers to evaluate the blue subtree

$$l = \max(l_1, l_2 + 1) = l_2 + 1$$

- If we evaluate the blue subtree first, we need l_2 registers to evaluate it, 1 register to hold its result and l_1 registers to evaluate the brown subtree

$$l = \max(l_1 + 1, l_2) = l_2$$





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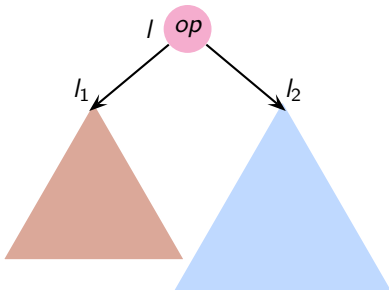
Assume that the register requirements of the two subtrees are l_1 and l_2 and that $l_1 < l_2$

- If we evaluate the brown subtree first, we need l_1 registers to evaluate it, 1 register to hold its result and l_2 registers to evaluate the blue subtree

$$l = \max(l_1, l_2 + 1) = l_2 + 1$$

- If we evaluate the blue subtree first, we need l_2 registers to evaluate it, 1 register to hold its result and l_1 registers to evaluate the brown subtree

$$l = \max(l_1 + 1, l_2) = l_2$$



Evaluate the subtree with larger requirements first



Labelling the Expression Tree

- For instruction with arity-2 and binary tree

$$label(n) = \begin{cases} 1 & n \text{ is a leaf and must be in a register} \\ 0 & n \text{ is a leaf and can be in memory} \\ \max(label(n_1), label(n_2)) & n \text{ has two child nodes } n_1 \text{ and } n_2 \\ & \text{and } label(n_1) \neq label(n_2) \\ label(n_1) + 1 & n \text{ has two child nodes } n_1 \text{ and } n_2 \\ & \text{and } label(n_1) = label(n_2) \end{cases}$$

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- For instruction with arity-2 and binary tree

$$label(n) = \begin{cases} 1 & n \text{ is a leaf and must be in a register} \\ 0 & n \text{ is a leaf and can be in memory} \\ \max(label(n_1), label(n_2)) & n \text{ has two child nodes } n_1 \text{ and } n_2 \\ & \text{and } label(n_1) \neq label(n_2) \\ label(n_1) + 1 & n \text{ has two child nodes } n_1 \text{ and } n_2 \\ & \text{and } label(n_1) = label(n_2) \end{cases}$$

- Generalizing to instructions and trees of higher arity

Let node n have k children with the labels $l_1 \geq l_2 \geq \dots \geq l_k$

$$label(n) = \max(l_j + j - 1), 1 \leq j \leq k$$



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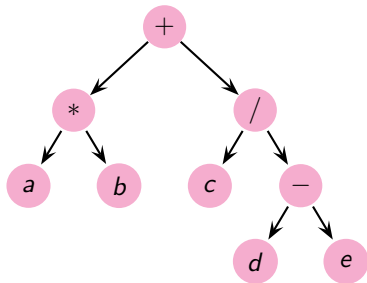
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Instruction Selection

Labelling the Expression Tree

Instructions

$$r \leftarrow r \text{ op } m$$
$$r_1 \leftarrow r_1 \text{ op } r_2$$
$$r \leftarrow m$$
$$m \leftarrow r$$




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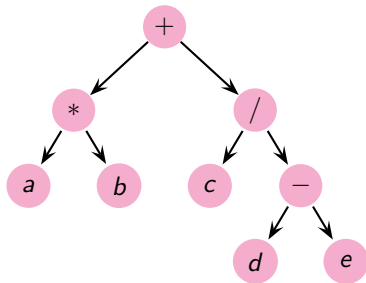
Managing Registers
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scip

Instruction Selection

Labelling the Expression Tree

Instructions

$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$


$$l_n = \begin{cases} 1 & n \text{ is a left leaf} \\ 0 & n \text{ is a right leaf} \\ \max(l_1, l_2) & n \text{ has two children with labels } l_1 \neq l_2 \\ l_1 + 1 & n \text{ has two children with labels } l_1 = l_2 \end{cases}$$



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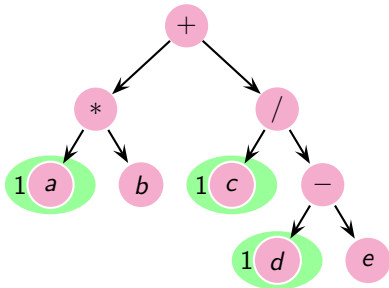
Instruction Selection

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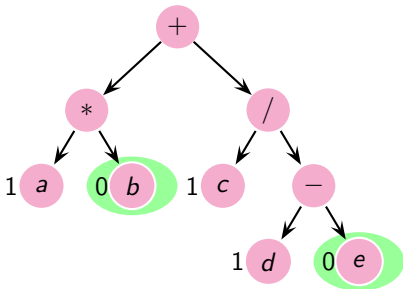
Instruction Selection

Labelling the Expression Tree

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$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$

$$l_n = \begin{cases} 1 & n \text{ is a left leaf} \\ 0 & n \text{ is a right leaf} \\ \max(l_1, l_2) & n \text{ has two children with labels } l_1 \neq l_2 \\ l_1 + 1 & n \text{ has two children with labels } l_1 = l_2 \end{cases}$$



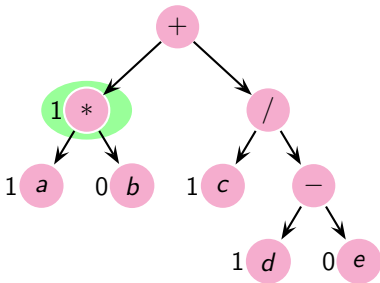


Labelling the Expression Tree

Instructions

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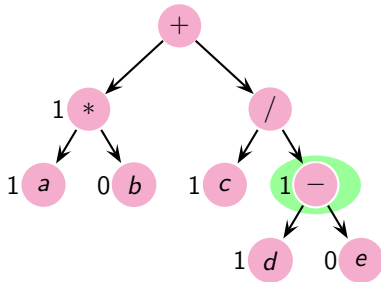


Labelling the Expression Tree

Instructions

$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$

$$l_n = \begin{cases} 1 & n \text{ is a left leaf} \\ 0 & n \text{ is a right leaf} \\ \max(l_1, l_2) & n \text{ has two children with labels } l_1 \neq l_2 \\ l_1 + 1 & n \text{ has two children with labels } l_1 = l_2 \end{cases}$$



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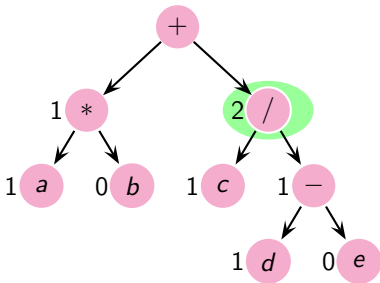
Instruction Selection

Labelling the Expression Tree

Instructions

$$\begin{aligned} r &\leftarrow r \text{ op } m \\ r_1 &\leftarrow r_1 \text{ op } r_2 \\ r &\leftarrow m \\ m &\leftarrow r \end{aligned}$$

$$l_n = \begin{cases} 1 & n \text{ is a left leaf} \\ 0 & n \text{ is a right leaf} \\ \max(l_1, l_2) & n \text{ has two children with} \\ & \text{labels } l_1 \neq l_2 \\ l_1 + 1 & n \text{ has two children with} \\ & \text{labels } l_1 = l_2 \end{cases}$$



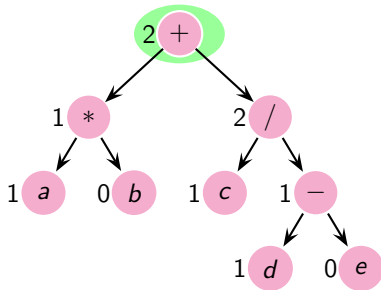


Labelling the Expression Tree

Instructions

```
 $r \leftarrow r \text{ op } m$   
 $r_1 \leftarrow r_1 \text{ op } r_2$   
 $r \leftarrow m$   
 $m \leftarrow r$ 
```

$$l_n = \begin{cases} 1 & n \text{ is a left leaf} \\ 0 & n \text{ is a right leaf} \\ \max(l_1, l_2) & n \text{ has two children with labels } l_1 \neq l_2 \\ l_1 + 1 & n \text{ has two children with labels } l_1 = l_2 \end{cases}$$



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Labelling the Expression Tree

Instructions

```

r ← r op m
r1 ← r1 op r2
r ← m
m ← r
    
```

Program 1

Regs r_0, r_1, r_2

```

r0 ← a
r0 ← r0 * b
    
```

```

r1 ← c
r2 ← d
r2 ← r2 - e
r1 ← r1 / r2
    
```

$r_0 \leftarrow r_0 + r_1$

Program 2

Regs r_0, r_1

```

r1 ← c
r0 ← d
r0 ← r0 - e
r1 ← r1 / r0
    
```

```

r0 ← a
r0 ← r0 * b
    
```

$r_0 \leftarrow r_0 + r_1$

Program 3

Regs r_0, r_1

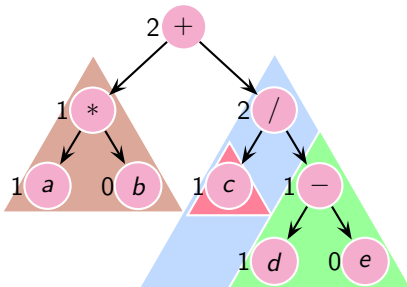
```

r0 ← d
r0 ← r0 - e
r1 ← c
r1 ← r1 / r0
    
```

```

r0 ← a
r0 ← r0 * b
    
```

$r_0 \leftarrow r_0 + r_1$



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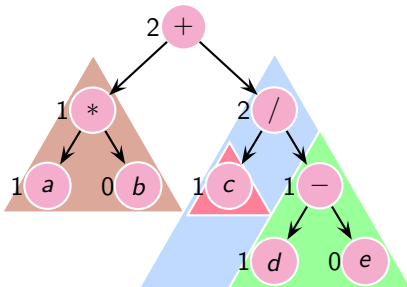
Instruction Selection

Labelling the Expression Tree

Instructions

```

r ← r op m
r1 ← r1 op r2
r ← m
m ← r
  
```



Program 1
Regs r_0, r_1, r_2

```

r0 ← a
r0 ← r0 * b
  
```

```

r1 ← c
r2 ← d
r2 ← r2 - e
r1 ← r1 / r2
  
```

$r_0 \leftarrow r_0 + r_1$

Program 2
Regs r_0, r_1

```

r1 ← c
r0 ← d
r0 ← r0 - e
r1 ← r1 / r0
  
```

```

r0 ← a
r0 ← r0 * b
  
```

$r_0 \leftarrow r_0 + r_1$

Program 3
Regs r_0, r_1

```

r0 ← d
r0 ← r0 - e
r1 ← c
r1 ← r1 / r0
  
```

```

r0 ← a
r0 ← r0 * b
  
```

$r_0 \leftarrow r_0 + r_1$

Program 1 computes the brown subtree first whereas programs 2 and 3 compute the blue subtree before the brown subtree; hence program 1 needs one register more than programs 2 and 3

The pink and the green subtrees have the same labels and the order does not matter; hence programs 2 and 3 have the same register requirements for the blue subtree



Generating Code for a Labelled Tree

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Instruction Selection

- Recursive procedure $gencode(n)$ to generate code for node n
 - $rstack$ is a stack of registers; $gencode(n)$ generates the code such that the result of the subtree rooted at n is contained in $top(rstack)$
 - $tstack$ is a stack of temporaries used when the algorithm runs out of registers
 - $swap(rstack)$ swaps the two top registers in $rstack$
 - Procedure $emit$ emits a single statement of the generated code

Generating Code for a Labelled Tree with k Registers



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Instruction Selection

- Procedure *gencode*(n) needs to consider the following five cases

1. n is a left leaf
2. The right child of n is a leaf
3. $\text{label}(\text{left child}) \geq \text{label}(\text{right child})$ and $\text{label}(\text{right child}) < k$
($\text{label}(\text{left child})$ can be $\geq k$)
4. $\text{label}(\text{left child}) < \text{label}(\text{right child})$ and $\text{label}(\text{left child}) < k$
($\text{label}(\text{right child})$ can be $\geq k$)
5. $\text{label}(\text{left child}) \geq k$ and $\text{label}(\text{right child}) \geq k$

Generating Code for a Labelled Tree with k Registers



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Instruction Selection

- Procedure *gencode*(n) needs to consider the following five cases
 1. n is a left leaf
 2. The right child of n is a leaf
 3. $\text{label}(\text{left child}) \geq \text{label}(\text{right child})$ and $\text{label}(\text{right child}) < k$
($\text{label}(\text{left child})$ can be $\geq k$)
 4. $\text{label}(\text{left child}) < \text{label}(\text{right child})$ and $\text{label}(\text{left child}) < k$
($\text{label}(\text{right child})$ can be $\geq k$)
 5. $\text{label}(\text{left child}) \geq k$ and $\text{label}(\text{right child}) \geq k$
- The above cases are exhaustive
 - Cases 1 and 2. Number k is irrelevant
 - Cases 2 and 3. At least one subtree has a label smaller than k
 - Case 5. The labels of both subtrees is greater than or equal to k

Generating Code for a Labelled Tree: Case 1



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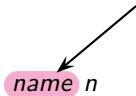
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Instruction Selection

Node n is a left leaf



Generating Code for a Labelled Tree: Case 1



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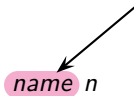
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Instruction Selection



Node n is a left leaf

Procedure $gencode(n)$ is

$emit(top(rstack) \leftarrow name)$

Generating Code for a Labelled Tree: Case 2



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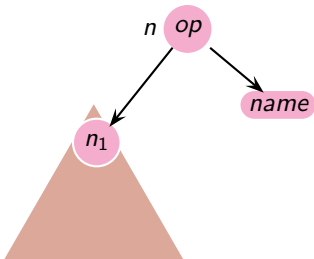
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Instruction Selection

The right child of n is a leaf



Generating Code for a Labelled Tree: Case 2



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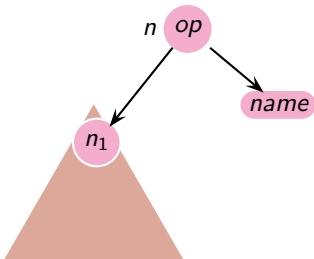
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Instruction Selection



The right child of n is a leaf

Procedure $gencode(n)$ is

```
 $gencode(n_1)$   
 $emit(top(rstack) \leftarrow top(rstack) \text{ op } name)$ 
```

Generating Code for a Labelled Tree: Case 3



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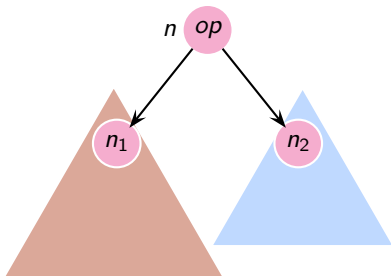
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Instruction Selection

$$\text{label}(n_1) \geq \text{label}(n_2) \text{ and } \text{label}(n_2) < k$$





Generating Code for a Labelled Tree: Case 3

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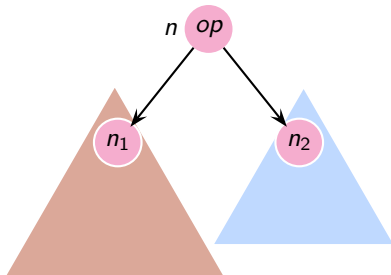
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Instruction Selection



$label(n_1) \geq label(n_2)$ and $label(n_2) < k$

Procedure $gencode(n)$ is

```
gencode( $n_1$ )  
 $R = pop(rstack)$   
gencode( $n_2$ )  
 $emit(R \leftarrow R \text{ op } top(rstack))$   
 $push(R, rstack)$ 
```

Contiguous evaluation guarantees that evaluation of n_2 is denied only a single register R

Generating Code for a Labelled Tree: Case 4



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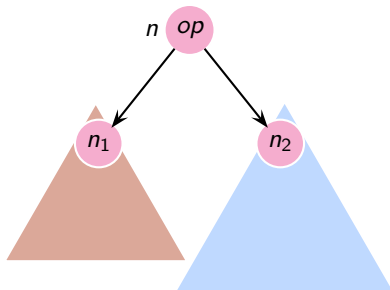
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Instruction Selection

$label(n_1) < label(n_2)$ and $label(n_1) < k$





Generating Code for a Labelled Tree: Case 4

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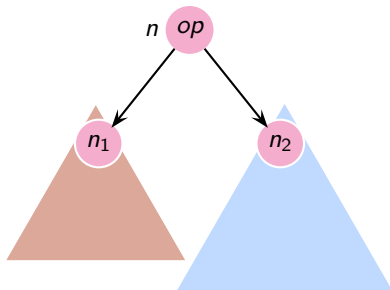
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Instruction Selection



$label(n_1) < label(n_2)$ and $label(n_1) < k$

Procedure $gencode(n)$ is

```
swap(rstack)
gencode(n2)
R = pop(rstack)
gencode(n1)
emit(top(rstack) ← top(rstack) op R)
push(R, rstack)
swap(rstack)
```

Contiguous evaluation guarantees that evaluation of n_1 is denied only a single register R



Generating Code for a Labelled Tree: Case 5

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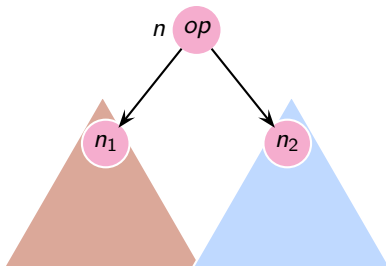
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Instruction Selection

$$\text{label}(n_1) \geq k \text{ and } \text{label}(n_2) \geq k$$





Generating Code for a Labelled Tree: Case 5

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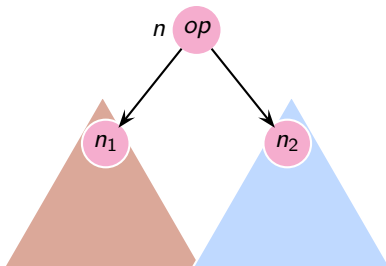
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Instruction Selection



$$label(n_1) \geq k \text{ and } label(n_2) \geq k$$

Procedure $gencode(n)$ is

```
gencode( $n_2$ )  
   $T = pop(tstack)$   
   $emit(T \leftarrow top(rstack))$   
  gencode( $n_1$ )  
   $emit(top(rstack) \leftarrow top(rstack) \text{ op } T)$   
   $push(T, tstack)$ 
```



Code Generation with Two Registers r_0 and r_1

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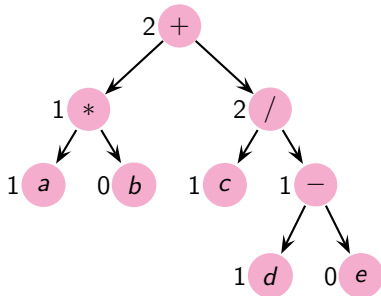
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Code Generation with Two Registers r_0 and r_1

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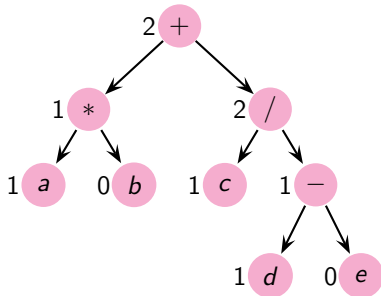
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Instruction Selection



We highlight the nodes in the expression tree
We show the control stack of invocations of
procedure *gencode* and highlight the state-
ments for cases 3 and 4

The red font shows the statements that are
being executed whereas blue font shows the
statements whose execution is over

C1	<i>emit</i> (<i>top</i> (<i>rstack</i>) $\leftarrow$ <i>name</i>)
C2	<i>gencode</i> (n_1) <i>emit</i> (<i>top</i> (<i>rstack</i>) $\leftarrow$ <i>top</i> (<i>rstack</i>) <i>op</i> <i>name</i>)
C3	<i>gencode</i> (n_1) <i>label</i> (<i>right</i>) $<$ <i>k</i> <i>R</i> = <i>pop</i> (<i>rstack</i>) <i>label</i> (<i>left</i>) $\geq$ <i>label</i> (<i>right</i>) <i>gencode</i> (n_2) <i>emit</i> (<i>R</i> $\leftarrow$ <i>R</i> <i>op</i> <i>top</i> (<i>rstack</i>)) <i>push</i> (<i>R</i> , <i>rstack</i>)
C4	<i>swap</i> (<i>rstack</i>) <i>label</i> (<i>left</i>) $<$ <i>k</i> <i>gencode</i> (n_2) <i>label</i> (<i>right</i>) $>$ <i>label</i> (<i>left</i>) <i>R</i> = <i>pop</i> (<i>rstack</i>) <i>gencode</i> (n_1) <i>emit</i> (<i>top</i> (<i>rstack</i>) $\leftarrow$ <i>top</i> (<i>rstack</i>) <i>op</i> <i>R</i>) <i>push</i> (<i>R</i> , <i>rstack</i>) <i>swap</i> (<i>rstack</i>)
C5	<i>gencode</i> (n_2) <i>T</i> = <i>pop</i> (<i>tstack</i>) <i>emit</i> (<i>T</i> $\leftarrow$ <i>top</i> (<i>rstack</i>)) <i>gencode</i> (n_1) <i>emit</i> (<i>top</i> (<i>rstack</i>) $\leftarrow$ <i>top</i> (<i>rstack</i>) <i>op</i> <i>T</i>) <i>push</i> (<i>T</i> , <i>tstack</i>)



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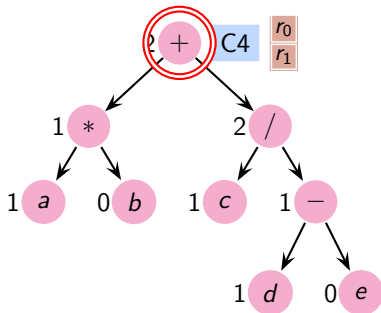
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Code Generation with Two Registers r_0 and r_1



$g(+): C4$

Control Stack

C1	<code>emit(top(rstack) ← name)</code>
C2	<code>gencode(n_1)</code> <code>emit(top(rstack) ← top(rstack) op name)</code> <code>gencode(n_1)</code> <code>label(right) < k</code> <code>R = pop(rstack)</code> <code>label(left) ≥ label(right)</code>
C3	<code>gencode(n_2)</code> <code>emit($R \leftarrow R$ op top(rstack))</code> <code>push(R, rstack)</code> <code>swap(rstack)</code> <code>label(left) < k</code> <code>gencode(n_2)</code> <code>label(right) > label(left)</code> <code>R = pop(rstack)</code>
C4	<code>gencode(n_1)</code> <code>emit(top(rstack) ← top(rstack) op R)</code> <code>push(R, rstack)</code> <code>swap(rstack)</code>
C5	<code>gencode(n_2)</code> <code>T = pop(tstack)</code> <code>emit($T \leftarrow top(rstack)$)</code> <code>gencode(n_1)</code> <code>emit(top(rstack) ← top(rstack) op T)</code> <code>push(T, tstack)</code>



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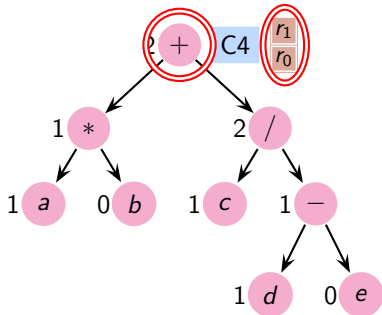
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Code Generation with Two Registers r_0 and r_1



$g(+): C4$

Control Stack

C1	<code>emit(top(rstack) ← name)</code>
C2	<code>gencode(n_1)</code> <code>emit(top(rstack) ← top(rstack) op name)</code>
C3	<code>gencode(n_1)</code> <code>label(right) < k</code> <code>R = pop(rstack)</code> <code>label(left) ≥ label(right)</code> <code>gencode(n_2)</code> <code>emit(R ← R op top(rstack))</code> <code>push(R, rstack)</code>
C4	<code>swap(rstack)</code> <code>label(left) < k</code> <code>gencode(n_2)</code> <code>label(right) > label(left)</code> <code>R = pop(rstack)</code> <code>gencode(n_1)</code> <code>emit(top(rstack) ← top(rstack) op R)</code> <code>push(R, rstack)</code> <code>swap(rstack)</code>
C5	<code>gencode(n_2)</code> <code>T = pop(tstack)</code> <code>emit(T ← top(rstack))</code> <code>gencode(n_1)</code> <code>emit(top(rstack) ← top(rstack) op T)</code> <code>push(T, tstack)</code>



Code Generation with Two Registers r_0 and r_1

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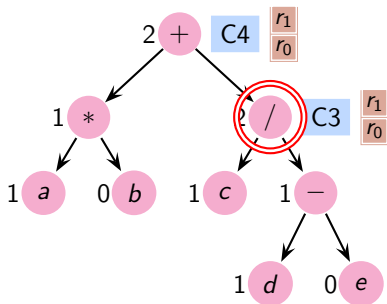
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$g(/) : C3$ $g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) \text{ op } name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R \text{ op } top(rstack))$ $push(R, rstack)$
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C5	$gencode(n_2)$ $T = pop(tstack)$ $emit(T \leftarrow top(rstack))$ $gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) \text{ op } T)$ $push(T, tstack)$



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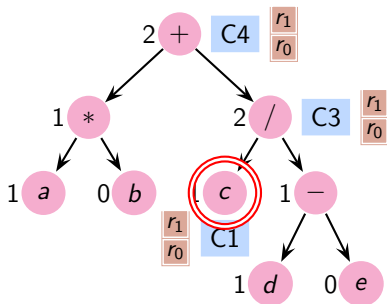
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Code Generation with Two Registers r_0 and r_1



$r_1 \leftarrow c$

$g(c) : C1$ $g(/) : C3$ $g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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C5	$gencode(n_2)$ $T = pop(tstack)$ $emit(T \leftarrow top(rstack))$ $gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op T)$ $push(T, tstack)$



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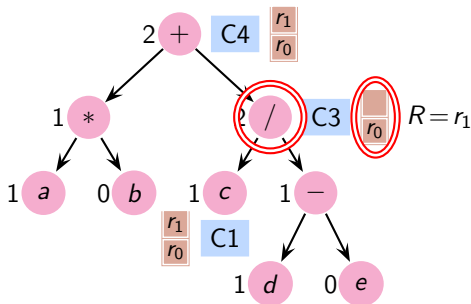
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Instruction Selection



$r_1 \leftarrow c$

$g(/) : C3$ $g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) \text{ op } name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R \text{ op } top(rstack))$ $push(R, rstack)$
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C5	$gencode(n_2)$ $T = pop(tstack)$ $emit(T \leftarrow top(rstack))$ $gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) \text{ op } T)$ $push(T, tstack)$



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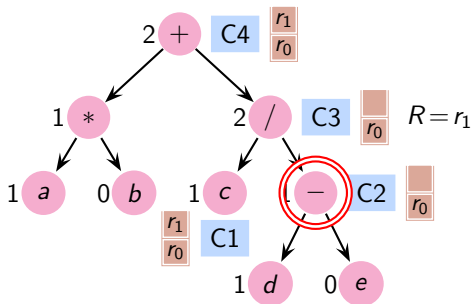
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Instruction Selection

Code Generation with Two Registers r_0 and r_1



$r_1 \leftarrow c$

$g(-) : C2$
$g(/) : C3$
$g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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Code Generation with Two Registers r_0 and r_1

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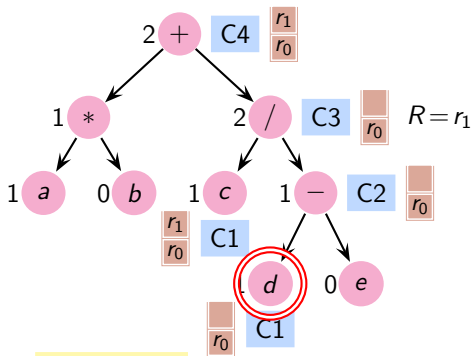
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$r_1 \leftarrow c$
 $r_0 \leftarrow d$

$g(d) : C1$
 $g(-) : C2$
 $g(/) : C3$
 $g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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C5	$gencode(n_2)$ $T = pop(tstack)$ $emit(T \leftarrow top(rstack))$ $gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op T)$ $push(T, tstack)$



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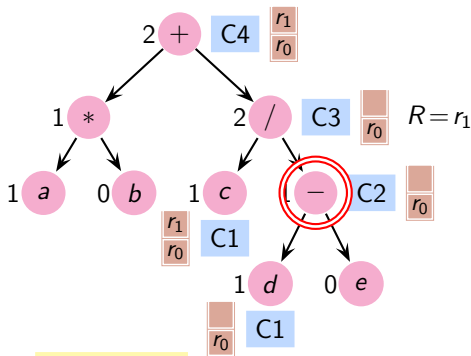
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$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$

$g(-) : C2$
 $g(/) : C3$
 $g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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Code Generation with Two Registers r_0 and r_1

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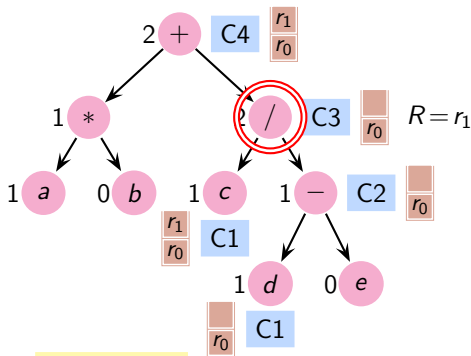
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$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$
 $r_1 \leftarrow r_1 / r_0$

$g(/) : C3$
 $g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
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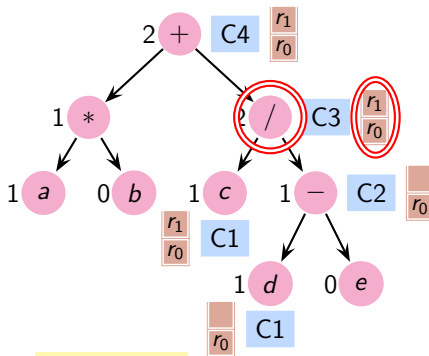
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Instruction Selection

Code Generation with Two Registers r_0 and r_1



$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$
 $r_1 \leftarrow r_1 / r_0$

$g(/) : C3$
 $g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
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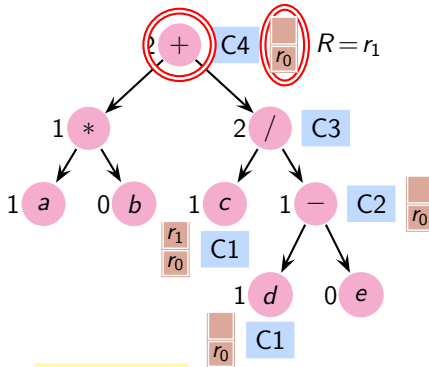
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Code Generation with Two Registers r_0 and r_1



$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$
 $r_1 \leftarrow r_1 / r_0$

$g(+): C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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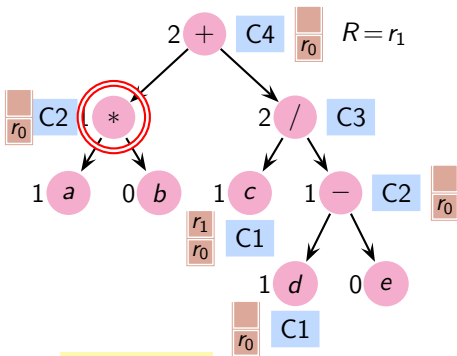
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Code Generation with Two Registers r_0 and r_1



$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$
 $r_1 \leftarrow r_1 / r_0$

$g(*) : C2$
 $g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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C5	$gencode(n_2)$ $T = pop(tstack)$ $emit(T \leftarrow top(rstack))$ $gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op T)$ $push(T, tstack)$



Code Generation with Two Registers r_0 and r_1

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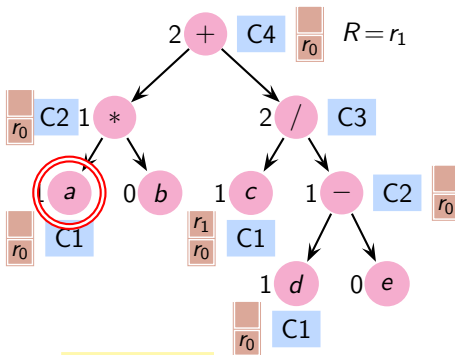
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$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$
 $r_1 \leftarrow r_1 / r_0$
 $r_0 \leftarrow a$

$g(a) : C1$
 $g(*) : C2$
 $g(+): C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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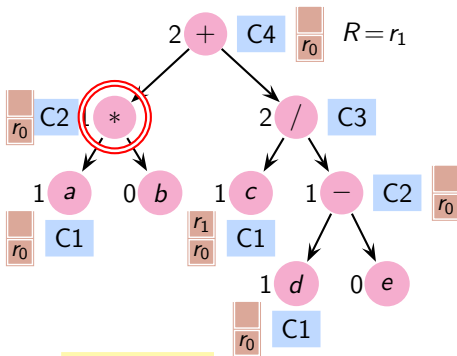
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Code Generation with Two Registers r_0 and r_1



$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$
 $r_1 \leftarrow r_1 / r_0$
 $r_0 \leftarrow a$
 $r_0 \leftarrow r_0 * b$

$g(*) : C2$
 $g(+) : C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$ $gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$
C3	$gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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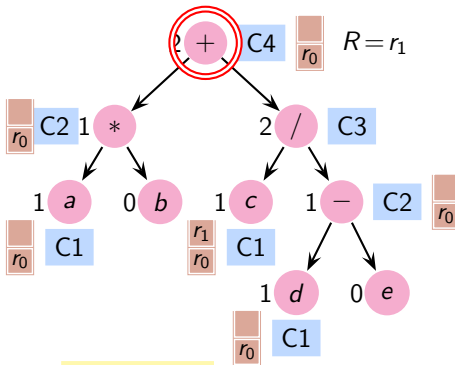
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Code Generation with Two Registers r_0 and r_1



$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$
 $r_1 \leftarrow r_1 / r_0$
 $r_0 \leftarrow a$
 $r_0 \leftarrow r_0 * b$
 $r_0 \leftarrow r_0 + r_1$

$g(+): C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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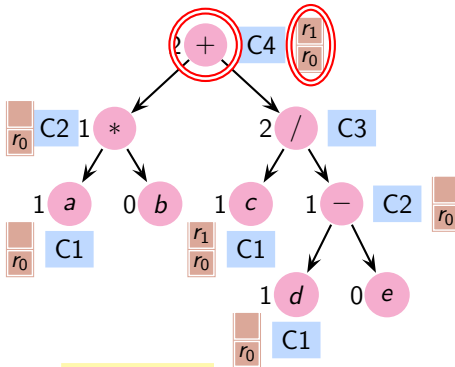
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Code Generation with Two Registers r_0 and r_1



$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$
 $r_1 \leftarrow r_1 / r_0$
 $r_0 \leftarrow a$
 $r_0 \leftarrow r_0 * b$
 $r_0 \leftarrow r_0 + r_1$

$g(+): C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
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C5	$gencode(n_2)$ $T = pop(tstack)$ $emit(T \leftarrow top(rstack))$ $gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op T)$ $push(T, tstack)$



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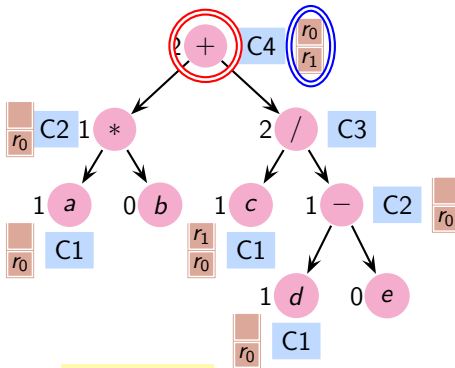
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Code Generation with Two Registers r_0 and r_1



$r_1 \leftarrow c$
 $r_0 \leftarrow d$
 $r_0 \leftarrow r_0 - e$
 $r_1 \leftarrow r_1 / r_0$
 $r_0 \leftarrow a$
 $r_0 \leftarrow r_0 * b$
 $r_0 \leftarrow r_0 + r_1$

$g(+): C4$

Control Stack

C1	$emit(top(rstack) \leftarrow name)$
C2	$gencode(n_1)$ $emit(top(rstack) \leftarrow top(rstack) op name)$
C3	$gencode(n_1)$ $label(right) < k$ $R = pop(rstack)$ $label(left) \geq label(right)$ $gencode(n_2)$ $emit(R \leftarrow R op top(rstack))$ $push(R, rstack)$
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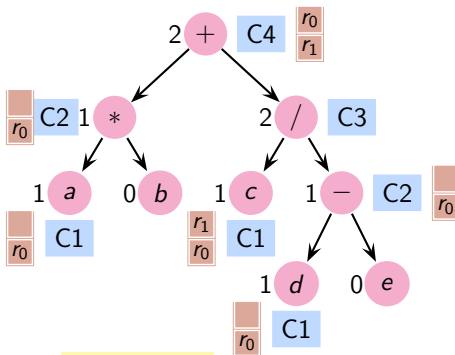
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Code Generation with Two Registers r_0 and r_1



```

 $r_1 \leftarrow c$ 
 $r_0 \leftarrow d$ 
 $r_0 \leftarrow r_0 - e$ 
 $r_1 \leftarrow r_1 / r_0$ 
 $r_0 \leftarrow a$ 
 $r_0 \leftarrow r_0 * b$ 
 $r_0 \leftarrow r_0 + r_1$ 

```

C1	$\text{emit}(\text{top}(\text{rstack}) \leftarrow \text{name})$
C2	$\text{gencode}(n_1)$ $\text{emit}(\text{top}(\text{rstack}) \leftarrow \text{top}(\text{rstack}) \text{ op } \text{name})$
C3	$\text{gencode}(n_1)$ $\text{label}(\text{right}) < k$ $R = \text{pop}(\text{rstack})$ $\text{label}(\text{left}) \geq \text{label}(\text{right})$ $\text{emit}(R \leftarrow R \text{ op } \text{top}(\text{rstack}))$ $\text{push}(R, \text{rstack})$
C4	$\text{swap}(\text{rstack})$ $\text{label}(\text{left}) < k$ $\text{gencode}(n_2)$ $\text{label}(\text{right}) > \text{label}(\text{left})$ $R = \text{pop}(\text{rstack})$ $\text{emit}(\text{top}(\text{rstack}) \leftarrow \text{top}(\text{rstack}) \text{ op } R)$ $\text{push}(R, \text{rstack})$ $\text{swap}(\text{rstack})$
C5	$\text{gencode}(n_2)$ $T = \text{pop}(\text{tstack})$ $\text{emit}(T \leftarrow \text{top}(\text{rstack}))$ $\text{gencode}(n_1)$ $\text{emit}(\text{top}(\text{rstack}) \leftarrow \text{top}(\text{rstack}) \text{ op } T)$ $\text{push}(T, \text{tstack})$



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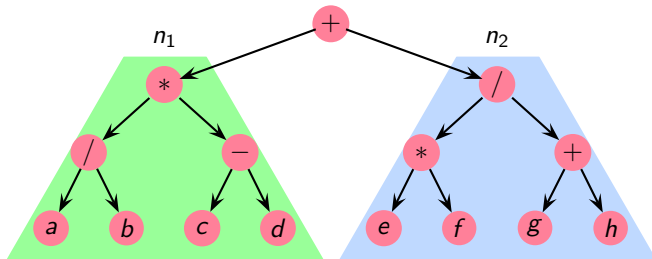
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Instruction Selection

Generate Code with Two Registers r_0 and r_1

Instructions

$$r \leftarrow r \text{ op } m$$
$$r_1 \leftarrow r_1 \text{ op } r_2$$
$$r \leftarrow m$$
$$m \leftarrow r$$




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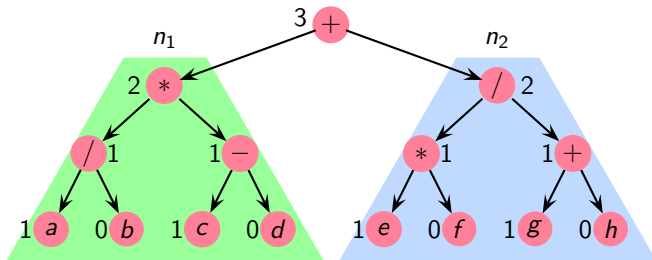
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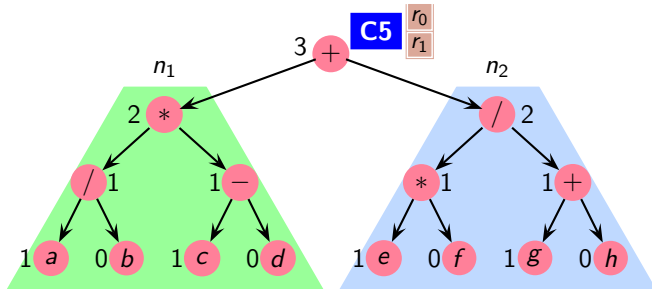
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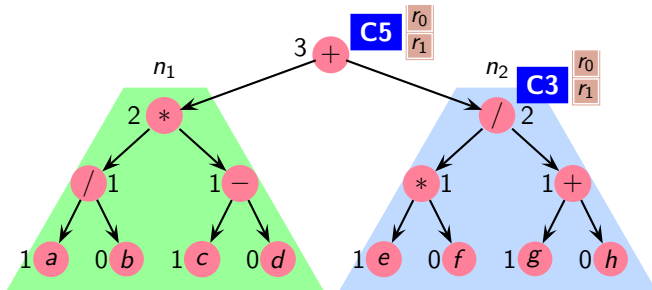
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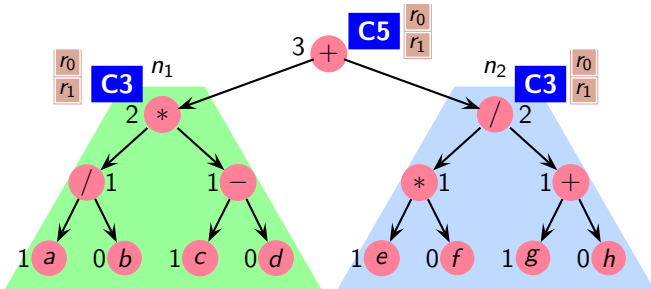
Instructions

$$r \leftarrow r \text{ op } m$$
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$$r \leftarrow m$$
$$m \leftarrow r$$
$$\begin{aligned} r_0 &\leftarrow e \\ r_0 &\leftarrow r_0 * f \\ r_1 &\leftarrow g \\ r_1 &\leftarrow r_1 + h \\ r_0 &\leftarrow r_0 / r_1 \\ t_0 &\leftarrow r_0 \end{aligned}$$




Generate Code with Two Registers r_0 and r_1

Instructions

$$r \leftarrow r \text{ op } m$$
$$r_1 \leftarrow r_1 \text{ op } r_2$$
$$r \leftarrow m$$
$$m \leftarrow r$$

$$r_0 \leftarrow e$$
$$r_0 \leftarrow r_0 * f$$
$$r_1 \leftarrow g$$
$$r_1 \leftarrow r_1 + h$$
$$r_0 \leftarrow r_0 / r_1$$
$$t_0 \leftarrow r_0$$
$$r_0 \leftarrow a$$
$$r_0 \leftarrow r_0 / b$$
$$r_1 \leftarrow C$$
$$r_1 \leftarrow r_1 - d$$
$$r_0 \leftarrow r_0 * r_1$$



Generate Code with Two Registers r_0 and r_1

Instructions

```

 $r \leftarrow r \text{ op } m$ 
 $r_1 \leftarrow r_1 \text{ op } r_2$ 
 $r \leftarrow m$ 
 $m \leftarrow r$ 

```

```

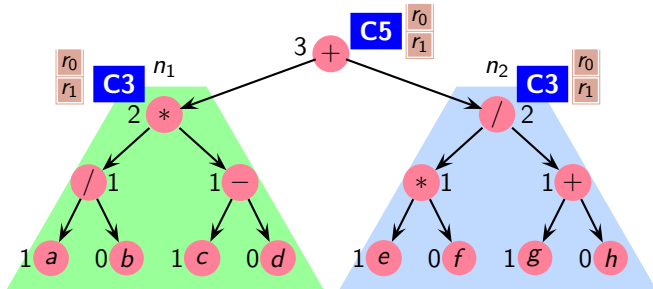
 $r_0 \leftarrow e$ 
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```

```

 $r_0 \leftarrow a$ 
 $r_0 \leftarrow r_0 / b$ 
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 $r_1 \leftarrow r_1 - d$ 
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```



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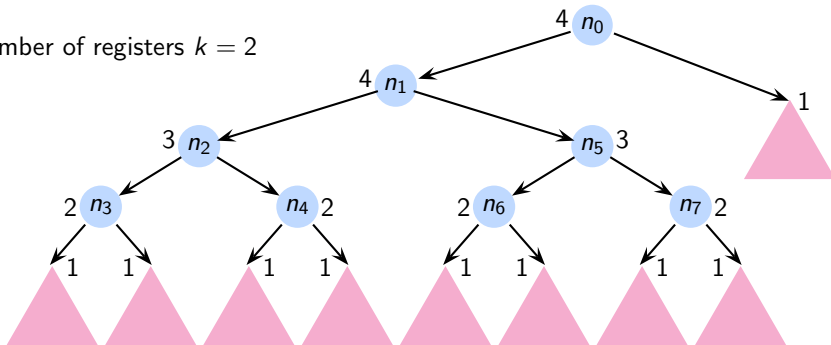
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Arguing Optimality: Dense Nodes and Major Nodes

Number of registers $k = 2$



We define node n to be a

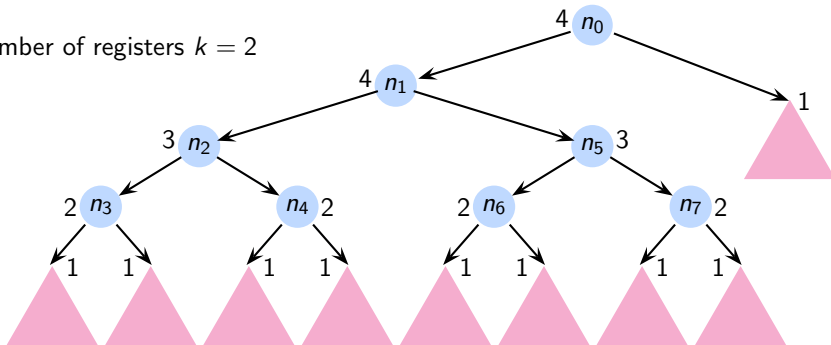
- *dense* node if $\text{label}(n) \geq k$
- *major* node if both its children are dense

A major node falls in case 5 of the
algorithm



Arguing Optimality: Dense Nodes and Major Nodes

Number of registers $k = 2$



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- *dense* node if $\text{label}(n) \geq k$
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A major node falls in case 5 of the algorithm

Dense Nodes = $\{n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7\}$
Major Nodes = $\{n_1, n_2, n_5\}$



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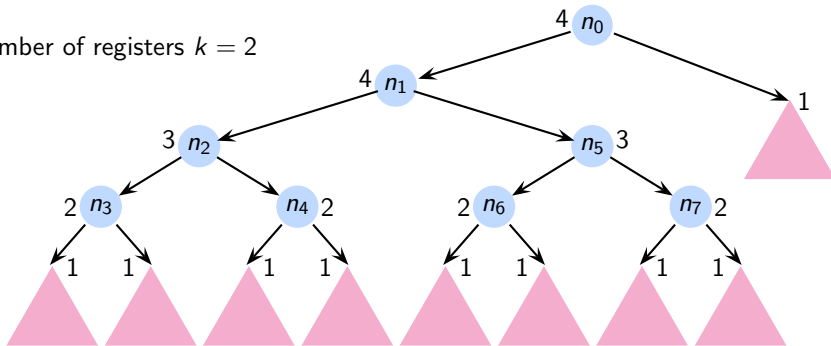
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Arguing Optimality: Dense Nodes and Major Nodes

Number of registers $k = 2$



- Every major node is dense but not vice-versa
- The parent of every dense node is dense but the parent of every major node need not be major (e.g., n_1 is major but n_0 is not)

Dense Nodes = $\{n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7\}$
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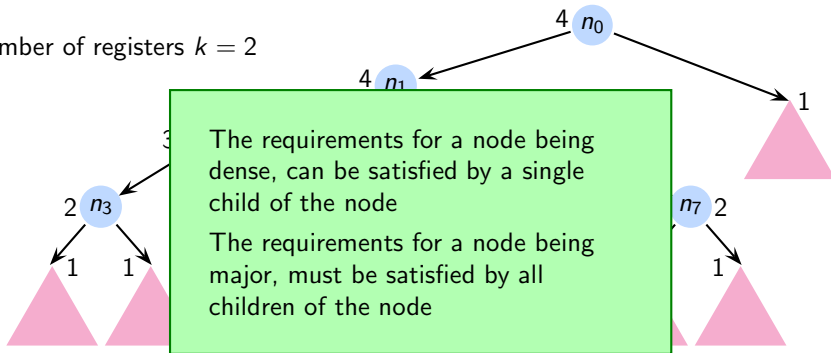
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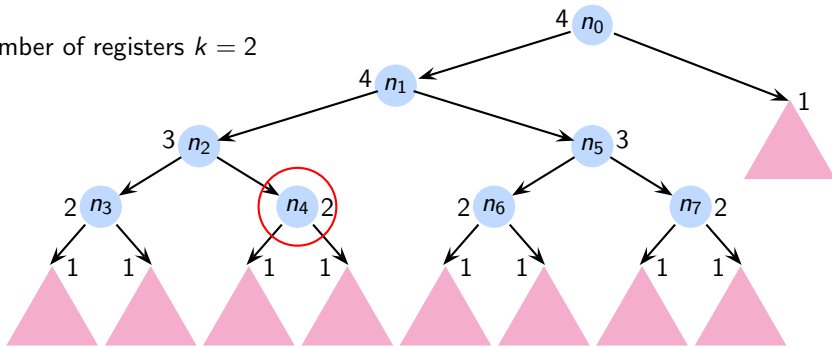
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Arguing Optimality: Dense Nodes and Major Nodes

Number of registers $k = 2$



If we store n_4 in memory,

Dense Nodes = $\{n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7\}$
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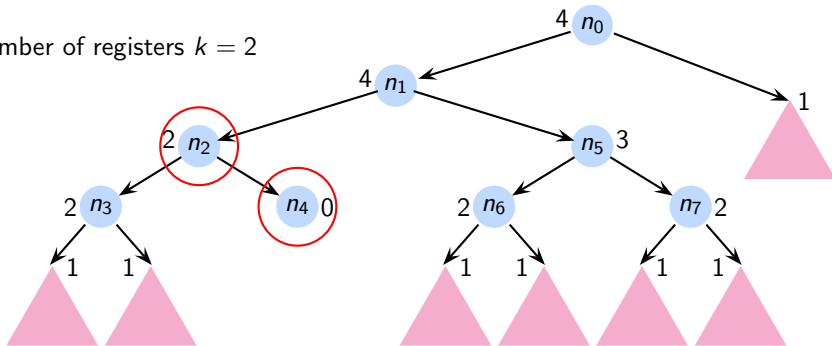
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Arguing Optimality: Dense Nodes and Major Nodes

Number of registers $k = 2$



If we store n_4 in memory,

- The $L(n_4)$ reduces to 0
- Hence, the label of n_2 reduces to 2. It remains dense but ceases to be major
- Node n_1 remains a major node

Dense Nodes = $\{n_0, n_1, n_2, n_3, n_5, n_6, n_7\}$
Major Nodes = $\{n_1, n_5\}$



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Optimality of the Sethi-Ullman Algorithm

- The algorithm generates
 - exactly one instruction per operator node (i.e., every internal node)
 - exactly one load per left leaf
 - no load for any right leaf



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Consider an expression tree with m major nodes



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Consider an expression tree with m major nodes

 - A store can reduce the number of major nodes by at most one



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Consider an expression tree with m major nodes

 - A store can reduce the number of major nodes by at most one
The node that becomes non-major, still remains a dense node so its parent remains a major node



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The node that becomes non-major, still remains a dense node so its parent remains a major node
 - Hence the tree would need at least m stores regardless of the algorithm used for generating code
 - The algorithm generates a single store for every major node (case 5), thus it generates exactly m stores



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Consider an expression tree with m major nodes

 - A store can reduce the number of major nodes by at most one
The node that becomes non-major, still remains a dense node so its parent remains a major node
 - Hence the tree would need at least m stores regardless of the algorithm used for generating code
 - The algorithm generates a single store for every major node (case 5), thus it generates exactly m stores
 - Since this is the smallest number of stores possible, the Sethi-Ullman algorithm is optimal